

A Dynamical Field Theory of Information and Attention

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Abstract

Conventionally, attention determines how available representations contribute to a query-dependent response. Neural architectures calculate this dependence through learned projections, query–context scores, regulation, and weighted aggregation. These separately executed operations construct one query-conditioned coupling.

Variational field theory determines the field response by varying an energy on admissible configurations coupled to an external source. Temporal development additionally requires a clock and a declared dynamical class.

Applied to information, available representations occupy a localized Hilbert carrier and primitive relations occupy a separate carrier. A closed relation operator maps relational amplitudes into compatible variations of the representations. Its range determines the active response sector, its adjoint gives relational field strength, and the squared norm of that strength determines the intrinsic energy up to positive calibration.

A query specified without access to its target changes the incident source without changing the field geometry. Compatible relational readings determine one query covector exactly when their differences lie in the null space of the relation operator. Orthogonal projection retains the component supported by admitted relations, and a declared readout produces the contextual output. This query-restricted variational response is attention. Query, key, and value are operational roles in the interaction rather than primitive field objects.

A supplied metric clock and the closed reversible variational class determine a second-order information-field law. Fixed-source initial data determine a unique global history in the energy space. Attention exposes the active response geometry, whereas causal response also resolves relational stiffness. Sources indistinguishable under every instantaneous query can therefore remain dynamically distinct.

Finite information fields admit exact digital and electromagnetic realizations. Digital realization evaluates the source-determined response directly in relation coordinates with explicit finite-precision error bounds. Electromagnetic realization intertwines the finite active dynamics with an invariant transverse sector of ideal Maxwell evolution. Both preserve the declared response law on distinct physical substrates.

Masked local-context observations from WikiText-2 determine a finite field source and relation-native incidents directly from their positive empirical measure. No question–answer supervision, learned parameters, gradient training, or pretrained representations enter this construction.

On finite associative retrieval, field response and FlashAttention both attain 1.000 accuracy across sequence lengths 64–512. Once relation coordinates and query, key, and value arrays have been formed, fused field response is 2.19–3.01 times faster for complete query batches and 3.13–3.64 times faster for incremental queries. Upstream neural representation learning is unnecessary for this task.

1 Introduction

1.1 Attention as a query-conditioned response

Attention determines how representations available before a query contribute to a response at the query site. Available representations establish possible distinctions and associations. A query specified without access to its target identifies which distinctions bear on the requested response. Admissibility and causal conditions exclude unavailable or future information, and a readout converts the resulting dependence into an output.

Neural architectures realize this duty through learned query, key, and value projections, pairwise scores, scaling, masking, normalization, and weighted summation [1, 9]. FlashAttention preserves scaled dot-product attention while reorganizing its evaluation to reduce memory traffic [3]. Linear-attention methods instead replace the pairwise score matrix by feature-space contractions or recurrent sufficient statistics [2, 5]. These constructions differ computationally but preserve one causal role. Previously available information regulates a later response according to the query.

Pairwise products, softmax, and weighted summation are sufficient implementations of that role, but they do not define it. Treating those operations as necessary would identify attention with one numerical realization. Structural redundancy means only that several separately executed operations realize one regulated query-conditioned coupling. No computational advantage follows from that observation without a separate implementation and measurement.

Information-field theory formulates the same coupling as a variational response. Relations admitted before the query determine a field geometry. A later query changes the incident source while leaving that geometry fixed. Variation of one source-coupled energy determines the regulated response. Attention is the query-induced part of that response rather than a geometric transcription of dot products or softmax.

1.2 Localized information and field geometry

Field theory separates field-defining data, configuration, external forcing, response, and temporal development. Applied to information, local quantities require a carrier and positive metric, while primitive relations require a distinct carrier and metric. A relation operator places relational amplitudes as compatible variations of the local quantities. Localization, positivity, closure, and overlap consistency must hold before these data determine a field.

A regular localized relation source is

$$R = (A, H, G, D), \quad D : \text{Dom}(D) \subseteq G \longrightarrow H.$$

Commutative algebra A records localization and restriction. Real Hilbert space H carries local quantity amplitudes, real Hilbert space G carries primitive relation amplitudes, and D is closed and densely defined. Carrier metrics determine the adjoint D^* .

A configuration $a \in \text{Dom}(D^*)$ assigns amplitudes to the quantity carrier. Relational field strength is $D^*a \in G$. Source geometry separates active and silent quantity directions through

$$B_R = \overline{\text{ran } D}, \quad N_R = \ker D^*, \quad P_R = P_{B_R}.$$

For a continuous incident-source covector $j \in H^*$, Riesz representative $j^\sharp \in H$, and positive stiffness calibration c , the source determines the intrinsic energy

$$V_R(a; j) = \frac{c}{2} \|D^*a\|_G^2 - j(P_R a).$$

Strong variation on $\text{Dom}(DD^*)$ gives

$$Y_R(a; j) = cDD^*a - P_R j^\sharp.$$

Weak variation remains defined on the full form domain. Consequently, field strength, energy, response sector, and static response all follow from one localized relation source before any query is presented.

1.3 Query-induced response

A query is a continuous localized covector $q \in H^*$ determined without access to its target response. It changes the incident source from j to $j + q$ while A , H , G , D , and the configuration a remain fixed. Its induced change in variational response is

$$\Delta_q Y_R(a; j) := Y_R(a; j + q) - Y_R(a; j) = -P_R q^\sharp.$$

Sign-adjusted readout by a bounded map $O : H \rightarrow Z$ defines

$$\text{Attn}_{R,O}(q) := -O \Delta_q Y_R(a; j) = O P_R q^\sharp.$$

Available relations determine B_R , query covector q selects the requested source direction, orthogonal projection excludes unsupported directions, and O selects the contextual output. Dependence on the query, dependence on admitted associations, masking, causal exclusion, and relabeling covariance follow from these objects. Several neural operations therefore realize duties supplied jointly by one projected variational response.

Observed input x may admit several compatible relation incidents $\mathcal{I}_R(x) \subseteq \text{Dom}(D)$. A faithful query covector exists exactly when

$$D\eta_1 = D\eta_2 \quad \text{for every } \eta_1, \eta_2 \in \mathcal{I}_R(x).$$

Canonicity therefore means that every compatible reading lies in one affine class modulo $\ker D$. A relation-native incident $[\eta] \in G/\ker D$ induces $q^\sharp = D\eta$, and attention reduces to $OD\eta$. Section 7 proves the response identity and its intervention consequences.

Energy-based and modern Hopfield accounts also connect attention with an energy or associative-memory update [8]. Present construction differs in what is determined before the query. Localization, carrier metrics, and primitive relations determine D , P_R , and the field energy, while the query appears only as an external-source covector. Query, key, and value consequently remain operational roles rather than primitive field objects.

1.4 Intrinsic dynamics and complete response

Static energy alone permits inequivalent temporal developments. Supplying a metric clock and restricting the admissible class to local, source-natural, closed reversible variational laws with positive energy and a field-independent second-order principal part determines, up to positive calibrations and temporal boundary terms,

$$m\ddot{a} + cDD^*a = P_R j^\sharp.$$

Fixed-source initial position and velocity determine a unique global history in the energy space. First-order realization couples quantity and relation variables through a skew-adjoint generator, while retarded response propagates admitted source histories causally.

Instantaneous attention exposes the active projector P_R . Complete causal response also exposes the intrinsic operator $L_R = cDD^*$ through later response derivatives and spectral residues. Two sources can therefore give the same attention response for every query while retaining different stiffness spectra and causal histories.

Learned state-space sequence models supply temporal recurrence through parameterized state transitions [4]. Information-field dynamics instead derive the intrinsic transition operator from D after a clock and the stated variational class are supplied. No absolute expressivity separation follows. A stateful attention realization given the same P_R , L_R , clock, incident map, readout, initial state, and temporal update reproduces the same finite causal response exactly.

Changing a query leaves D fixed. Changing an admitted relation replaces D and redetermines the active sector, energy, intrinsic operator, and subsequent response. Exact addition, removal, smooth transport, and rank-transition identities distinguish source variation from evolution on a fixed source.

1.5 Evidence, principal results, and organization

Table 1: WikiText-2 source construction and held-out response from directed local occurrences with their positive empirical measure. Training text alone determines D ; no question–answer supervision, learned parameters, gradient training, or pretrained representations enter source construction or response.

Quantity	Value
Held-out occurrences	2,000
Target coverage	99.25%
Frequency top-1 / top-5	10.15% / 28.30%
Relational top-1 / top-5	15.75% / 34.25%
Mean reciprocal rank	0.2539

Table 2: Relation-coordinate response and one direct FP16 FlashAttention call on the associative-retrieval experiment. Both realizations attain 1.000 retrieval accuracy. Latency ratios begin after relation coordinates and query, key, and value arrays have been formed.

Sequence / bindings	64 / 4	128 / 8	256 / 16	512 / 64
Complete-query speedup	2.59×	2.19×	2.42×	3.01×
Incremental-query speedup	3.64×	3.13×	3.36×	3.23×

Finite information fields also admit exact digital and electromagnetic realizations. Digital realization evaluates source-determined operators directly in relation coordinates and through response-preserving temporal reductions. Electromagnetic realization intertwines the finite active law with an invariant transverse sector of ideal Maxwell evolution, preserving mass, stiffness, source coupling, energy, and retarded response. These realizations leave the information field independent of its computational or physical substrate.

Principal results follow their logical dependence.

1. *Source and geometry.* Regular localized quantities, positive carrier metrics, and primitive relations determine D , the active and silent sectors, and the intrinsic operator. Admitted

observations determine such a source only relative to a stated source class and observation map.

2. *Energy and attention.* Carrier isotropy and minimal relational order determine the quadratic intrinsic energy up to positive calibration. Query-induced variation gives $\Delta_q Y_R = -P_R q^\sharp$, and sign-adjusted readout gives $OP_R q^\sharp$.
3. *Intrinsic dynamics.* A supplied metric clock and the closed reversible variational class determine the second-order field equation. Fixed-source Cauchy data give a unique global energy-space history.
4. *Complete response and source variation.* Response jets, reciprocity, spectral residues, and first-order quantity–relation exchange expose structure absent from instantaneous attention. Exact source-variation identities determine how changing relations alter subsequent response.
5. *Finite realizations.* Corpus, relation-coordinate, state-space, and electromagnetic constructions realize the same declared field law in distinct observational, digital, and physical settings. Computational comparisons report both their mathematical preconditions and measured execution costs.

Sections 2–6 distinguish source data from configurations and derive the localized source, field geometry, energy, and static response. Sections 7–8 establish attention and source variation. Sections 9–14 develop intrinsic dynamics, first-order exchange, causal response, time-dependent geometry, and local differential realization. Sections 15–18 give finite examples and corpus, digital, and electromagnetic realizations. Sections 19–20 discuss the resulting distinctions and conclude.

2 Source data, field configurations, and response laws

2.1 Supplied, selected, and derived objects

Information-field determination uses objects with distinct mathematical roles. Recorded observations constrain admissible sources. A selected source determines field geometry and a response law. Configuration, incident source, clock, initial data, and readout remain separate inputs. Identifying any two of these objects without a map between them changes the claim.

Recorded observations have force only relative to a declared observation carrier, candidate source class, and admission rule. Equal byte strings or equal numerical arrays can support different source assignments when their operational roles differ. Section 4 defines the compatible source family and its non-identifiability boundary.

Localized source data precede every field configuration. Algebra A records localization and restriction, H carries quantity amplitudes, G carries primitive relation amplitudes, and D places relation amplitudes as compatible quantity variations. Carrier metrics belong to the source because they define adjoints, norms, and orthogonality.

Field geometry is derived rather than independently supplied. In particular,

$$B_R = \overline{\text{ran } D}, \quad N_R = \ker D^*, \quad H = B_R \oplus N_R.$$

Configuration a and incident source j select one evaluation of the response law. Strong response is represented by an element of H when $a \in \text{Dom}(DD^*)$. Weak response at finite energy is instead a continuous functional on the form domain $\text{Dom}(D^*)$. Section 6 derives both forms.

Clock, initial data, source history, and readout remain external to static source geometry. Repeated evaluation of a static response does not supply temporal development, and changing a readout does not change the response being observed.

Table 3: Mathematical roles in information-field determination.

Object	Status	Role
Recorded observations	Supplied	Restrict a declared candidate source class through an admission rule.
Localized source R	Selected or determined	Supplies localization, carrier metrics, quantity and relation carriers, and D .
Field geometry	Derived	Contains B_R , N_R , P_R , D^* , and $L_R = cDD^*$.
Configuration a	Selected	Assigns quantity amplitudes in $\text{Dom}(D^*)$.
Incident source j	Supplied	Couples a probe or forcing to a fixed field.
Static response	Derived	Evaluates the variation at one configuration and incident source.
Clock and initial data	Supplied	Specify temporal comparison and Cauchy data.
History	Derived	Solves the temporal equation for the supplied data and source history.
Readout O	Supplied	Selects an observable component of a configuration, response, or history.

2.2 Order of determination

Logical dependence begins with a declared candidate source class $\mathfrak{S}$, an observation carrier, and an admission rule. Recorded observations select a compatible subfamily of $\mathfrak{S}$. An empty family represents inconsistency, a singleton up to source isomorphism represents determination, and a larger family represents unresolved source ambiguity. Choosing one member of a larger family supplies additional data.

A selected regular source determines its field geometry and static response law. Local relation equations act on their admitted regions, while shared quantities and interface conditions couple those equations. Compatibility and gluing produce the global relation operator and active sector under the hypotheses of Sections 3 and 5.

For fixed R , selecting $a \in \text{Dom}(D^*)$ and $j \in H^*$ evaluates the already determined response law. A query is a distinguished admissible increment of j . Query application therefore changes an incident coordinate without constructing the field or modifying its primitive relations.

Temporal determination follows only after a metric clock, initial position, initial velocity, and source history are supplied. Within the intrinsic closed reversible variational class, these data determine the temporal equation and one realized history under the well-posedness hypotheses of Section 9.

This dependence is logical rather than chronological. Observations may be gathered over time, and later evidence may select a new source. Every source revision starts a new determination unless an exact update theorem transports the derived geometry.

Restriction of an already determined field also differs from redetermination using a truncated source. Restriction observes part of the original response, whereas truncation deletes source data before geometry is derived. Equality requires the locality and gluing hypotheses proved later rather than following from notation alone.

2.3 Response values and response laws

Configuration a is one admissible assignment of quantity amplitudes. Realized response $Y_R(a; j)$ is the value produced at one configuration under one incident source. Complete static response law is the map

$$(a, j) \mapsto Y_R(a; j)$$

over every admitted configuration and incident source. A temporal law is instead a differential equation on histories. Agreement of two realized responses at one pair (a_0, j_0) establishes none of the corresponding equalities between response laws or temporal laws.

A zero observed value can arise from distinct structures.

Table 4: Inequivalent causes of a zero observed response.

Cause	Mathematical location
Response channel is absent	Source type or incident class
Incident amplitude is zero	Incident source
Nonzero incident is projected out	Active-sector geometry
Configuration and forcing balance	Realized response
Nonzero response lies in $\ker O$	Readout

Equal observed zeros therefore do not identify channel existence, excitation, active support, equilibrium, or observability. Equality of $Y_R(a_0; j_0)$ at one pair likewise does not imply equality of P_R , L_R , or the complete response law. Response jets and spectral data in Section 11 distinguish fields that one realized value cannot.

Static response law also differs from temporal development. Map Y_R gives a variational covector at one configuration. A temporal equation relates configuration and velocity across clock values. Several temporal laws can share one static energy until additional admissibility conditions select the intrinsic inertial law.

2.4 Fixed-source and source-changing variations

Input type is determined by which object an input is permitted to change.

Table 5: Effects of the input classes used in the field theory.

Input	Changes	Preserves
Query q	Incident source $j \mapsto j + q$	Source, geometry, configuration, and readout
Configuration variation	Configuration a	Source, response law, incident source, and readout
Readout replacement	Observation map O	Source, configuration, and field response
Clock recalibration	Temporal coordinate and calibrated coefficients	Static source geometry and static response law
Structural intervention	Source $R \mapsto R'$	No derived geometry without an update theorem

For fixed R , background incident source j , and query covector q , query application gives

$$Y_R(a; j + q) - Y_R(a; j) = -P_R q^\sharp$$

while A , H , G , D , D^* , P_R , and L_R remain fixed. Relation-native incidents have intrinsic type

$$[\eta] \in \text{Dom}(D)/\ker D, \quad q^\sharp = D\eta \in B_R.$$

Finite and bounded branches satisfy $\text{Dom}(D) = G$, reducing this quotient to $G/\ker D$. Identity $P_R D\eta = D\eta$ then removes any further projector after source admission.

Configuration variation replaces a by $a + \varepsilon\xi$ while preserving the source and probes intrinsic stiffness. Readout replacement changes only the observed component. Clock recalibration changes temporal representation and calibrated coefficients without changing static source geometry.

A structural intervention acts instead as a partial map on the source class,

$$\sigma : \mathfrak{S} \rightarrow \mathfrak{S}, \quad R \mapsto R'.$$

Derived objects must then be redetermined from R' , including D'^* , $B_{R'}$, $N_{R'}$, $P_{R'}$, and $L_{R'}$. Exact update and downdate identities can reduce the required computation without changing this logical dependence.

These distinctions fix the role of every subsequent input. [Section 3](#) defines the localized relation source and its analytic hypotheses.

3 Localized relation sources

3.1 Localization and carrier data

A localized relation source records the information available before any field configuration, incident source, or query is chosen. Localization distinguishes regions and their overlaps. Quantity variables assign amplitudes to localized informational sites. Relation variables assign amplitudes to primitive compatibility relations. A relation operator places those relational amplitudes as compatible variations of the quantities.

Localization is encoded by a commutative real C^* -algebra A , a region family Λ_R , and nondegenerate $*$ -representations

$$\pi_H : A \longrightarrow \mathcal{B}(H), \quad \pi_G : A \longrightarrow \mathcal{B}(G)$$

on real Hilbert spaces H and G . Every $U \in \Lambda_R$ specifies a localization ideal $A(U) \subseteq A$, together with its inclusions and overlaps. Corresponding regional carriers are

$$H(U) = \overline{\pi_H(A(U))H}, \quad G(U) = \overline{\pi_G(A(U))G}.$$

Quantity and relation carriers have distinct roles. A quantity configuration lies in H , a primitive relation amplitude lies in G , and a densely defined operator

$$D : \text{Dom}(D) \subseteq G \longrightarrow H$$

relates them. Positive inner products on H and G determine lengths, adjoints, and orthogonality. Identical incidence coefficients equipped with different carrier metrics therefore define different sources. No canonical Euclidean metric follows from a coordinate array.

Neither carrier is identified with its coordinate representation. Array order, token order, memory address, and processor topology define no neighborhood unless the localization datum assigns them that role. Regional dependence is carried by A , its represented ideals, and the support of D . Queries, field configurations, initial data, clocks, and readouts do not belong to the source.

3.2 Localized relation sources and isomorphisms

Definition 3.1 (Localized relation source). A localized relation source is a quadruple

$$R = (A, H, G, D)$$

with the following data.

1. A is a commutative real C^* -algebra equipped with a region family Λ_R and nondegenerate representations π_H and π_G .
2. H is the real Hilbert carrier of quantity configurations.
3. G is the real Hilbert carrier of primitive relation amplitudes.
4. $D : \text{Dom}(D) \subseteq G \rightarrow H$ is densely defined and closed.
5. Regional carriers, restriction maps, local operator domains, and interface data satisfy the compatibility and gluing conditions of [section 3.3](#).

Notation $R = (A, H, G, D)$ suppresses Λ_R , π_H , and π_G while retaining them as source data. Such suppression never identifies the represented localization datum with the bare algebra.

A source isomorphism $\Phi : R \rightarrow R'$ consists of an algebra isomorphism $\alpha : A \rightarrow A'$, a region bijection $\phi_\Lambda : \Lambda_R \rightarrow \Lambda_{R'}$, and Hilbert-space isometries

$$U_H : H \longrightarrow H', \quad U_G : G \longrightarrow G'.$$

Localization is preserved when

$$\alpha(A(U)) = A'(\phi_\Lambda(U)), \quad U \in \Lambda_R,$$

and

$$U_H \pi_H(f) = \pi'_H(\alpha f) U_H, \quad U_G \pi_G(f) = \pi'_G(\alpha f) U_G, \quad f \in A.$$

Relation structure is preserved when

$$U_G \text{Dom}(D) = \text{Dom}(D'), \quad U_H D \eta = D' U_G \eta, \quad \eta \in \text{Dom}(D).$$

Relabeling, coordinate permutation, and orthonormal basis changes therefore preserve a source only when they transport localization, both carrier metrics, and the relation operator together.

3.3 Analytic assumptions and regional compatibility

Subsequent results use three nested analytic cases.

Finite case. Both carriers are finite-dimensional and $\text{Dom}(D) = G$. Every linear map is bounded, and every range is closed.

Bounded Hilbert case. Both carriers may be infinite-dimensional and $D \in \mathcal{B}(G, H)$. Adjoint $D^* \in \mathcal{B}(H, G)$ is defined on all of H , although $\text{ran } D$ need not be closed.

Closed-operator case. Both carriers may be infinite-dimensional and D is densely defined and closed. Hilbert adjoint D^* is densely defined and closed, while DD^* is nonnegative and self-adjoint on

$$\text{Dom}(DD^*) = \{a \in \text{Dom}(D^*) : D^*a \in \text{Dom}(D)\}.$$

Energy is defined on the form domain $\text{Dom}(D^*)$. Equipped with the form norm

$$\|a\|_{D^*}^2 = \|a\|_H^2 + \|D^*a\|_G^2,$$

this domain carries the weak static response as a continuous functional. A strong H -valued response requires $a \in \text{Dom}(DD^*)$. [Appendix A](#) identifies this strong operator domain and represents weak response on the full form domain.

No closed-range assumption is needed to define

$$B_R = \overline{\text{ran } D}$$

or its orthogonal projector. Closed range is imposed only when a result uses an inverse on B_R , a bounded Moore–Penrose inverse, uniform transport under source variation, or an estimate requiring a spectral gap at zero.

Regional data obey four compatibility conditions. For each inclusion $V \subseteq U$, bounded restriction maps

$$\rho_{VU}^H : H(U) \longrightarrow H(V), \quad \rho_{VU}^G : G(U) \longrightarrow G(V)$$

satisfy identity and composition laws, while regional metrics agree with the restrictions of the carrier metrics. Assignments agreeing after restriction to every overlap glue uniquely on each declared covering family. Every local term of D has declared regional support and a compatible local domain. A regional operator equation is asserted only when its region contains the complete support of every term involved or when the missing interface values are supplied as boundary data.

Compatible local actions assemble to the densely defined closed operator D . Abstract Hilbert spaces alone provide none of this localization or gluing structure. Nothing in these conditions identifies the field redetermined from a truncated source $R|_U$ with the restriction to U of the field determined by R ; such an equality requires a separate locality theorem.

3.4 Finite weighted-incidence realization

Let $\mathbf{X}$ be a finite set of quantity sites and $\mathbf{E}$ a finite set of primitive relations. Positive weights

$$\mu_p > 0, \quad p \in \mathbf{X}, \quad w_r > 0, \quad r \in \mathbf{E}$$

define

$$\begin{aligned} H &= \mathbb{R}^{\mathbf{X}}, & \langle a, b \rangle_H &= \sum_{p \in \mathbf{X}} \mu_p a_p b_p, \\ G &= \mathbb{R}^{\mathbf{E}}, & \langle \eta, \zeta \rangle_G &= \sum_{r \in \mathbf{E}} w_r \eta_r \zeta_r. \end{aligned}$$

Every relation r supplies a finitely supported section

$$b_r = \sum_{p \in \mathbf{X}} b_{pr} e_p^H \in H,$$

and D is determined on the relation basis by

$$D e_r^G = b_r.$$

Writing

$$M_H = \text{diag}(\mu_p), \quad W_G = \text{diag}(w_r),$$

gives

$$D^* = W_G^{-1} D^\top M_H, \quad (D^* a)_r = \frac{1}{w_r} \sum_{p \in \mathbf{X}} \mu_p b_{pr} a_p.$$

Finite localization uses the pointwise algebra

$$A = \mathbb{R}^{\mathbf{X} \sqcup \mathbf{E}}.$$

Characteristic functions of subsets are localization idempotents. Coordinate representations on H and G retain the quantity and relation coordinates, respectively. Support of b_r records the quantity sites incident to relation r .

No key-payload partition is required by [definition 3.1](#). Such a partition is additional data in particular attention realizations. Relation sections may have arbitrary declared finite support and coefficients. Duplicate or linearly dependent sections can change D and DD^* without changing $\text{ran } D$.

Proposition 3.2 (Finite weighted adjoint). *Positive carrier weights and relation coefficients uniquely determine*

$$D^* = W_G^{-1} D^\top M_H.$$

Proof. For arbitrary $\eta \in G$ and $a \in H$,

$$\langle D\eta, a \rangle_H = \eta^\top D^\top M_H a = \eta^\top W_G (W_G^{-1} D^\top M_H a).$$

Last expression equals $\langle \eta, W_G^{-1} D^\top M_H a \rangle_G$, which determines the adjoint uniquely. $\square$

3.5 Restrictions, boundary data, and gluing

Finite localization makes regional compatibility explicit. A region $U \subseteq \mathbf{X} \sqcup \mathbf{E}$ carries

$$H(U) = \text{span}\{e_p^H : p \in U \cap \mathbf{X}\}, \quad G(U) = \text{span}\{e_r^G : r \in U \cap \mathbf{E}\}.$$

Restriction deletes coordinates outside U . A relation $r \in U \cap \mathbf{E}$ whose section b_r meets a quantity site outside U has incomplete support at the boundary. Evaluation of that relation requires the missing quantity values as boundary data or a larger region containing $\text{supp } b_r$.

Proposition 3.3 (Finite gluing). *Let $\{U_i\}_{i \in \mathcal{K}}$ cover $\mathbf{X} \sqcup \mathbf{E}$. Suppose local assignments*

$$(a_i, \eta_i) \in H(U_i) \oplus G(U_i)$$

agree after restriction to every overlap. A unique global assignment

$$(a, \eta) \in H \oplus G$$

restricts to (a_i, η_i) on each U_i .

Proof. Every quantity or relation coordinate belongs to at least one region. Overlap agreement makes its value independent of the selected region. Coordinatewise values define one global pair, and any other pair with the same restrictions agrees on every coordinate. $\square$

Operator assembly requires every primitive term to occur with complete support. Suppose each relation section b_r has $\{r\} \cup \text{supp } b_r \subseteq U_i$ for at least one i . Overlap agreement then assigns one amplitude to r and one value to every incident quantity site. Summing the resulting terms $\eta_r b_r$ gives the unique global action

$$D\eta = \sum_{r \in \mathbf{E}} \eta_r b_r.$$

Compatible local equations on such a cover therefore imply the assembled global equation. Regions omitting part of a relation support require the omitted values as boundary data and cannot independently determine that term.

Source data and the conditions under which their local terms assemble into one relation operator are therefore fixed. [Section 4](#) determines when recorded observations select one such source rather than a nontrivial compatible family.

4 Source determination from observations

4.1 Observation maps and constraints

Observed records restrict a declared source class only through a declared observation map. Fix a class $\mathfrak{S}$ of localized relation sources, a class $\mathcal{U}$ of observation interventions, an observation carrier $\mathcal{Y}$, and a map

$$h : \mathfrak{S} \times \mathcal{U} \longrightarrow \mathcal{Y}.$$

Intervention $\iota \in \mathcal{U}$ specifies what was applied, withheld, or measured. Observation carrier $\mathcal{Y}$ specifies the type of the resulting record. Equal strings or coordinate arrays in different carriers constitute equal observations only when a declared map identifies their roles.

An observation constraint is a pair

$$o = (\iota_o, C_o), \quad \iota_o \in \mathcal{U}, \quad C_o \subseteq \mathcal{Y}.$$

Exact observation $y_o \in \mathcal{Y}$ uses $C_o = \{y_o\}$. Set-valued or finite-precision evidence may use a larger admissible set. No probability measure, metric, or likelihood on $\mathcal{Y}$ is required for this compatibility relation.

Source determination is conditional on $\mathfrak{S}$, $\mathcal{U}$, $\mathcal{Y}$, and h . Recorded marks alone supply none of these mathematical roles.

4.2 Compatible source families

One constraint selects the compatible source family

$$\mathcal{C}(o) = \{R \in \mathfrak{S} : h(R, \iota_o) \in C_o\}.$$

For a finite collection $\mathfrak{D}$ of constraints, define

$$\mathcal{A}(\mathfrak{D}) = \bigcap_{o \in \mathfrak{D}} \mathcal{C}(o).$$

Three outcomes remain distinct.

$$\begin{aligned} \mathcal{A}(\mathfrak{D}) = \emptyset &\implies \text{inconsistent observations relative to } \mathfrak{S}, \\ \mathcal{A}(\mathfrak{D})/\cong = \{[R]\} &\implies \text{source determined up to isomorphism,} \\ |\mathcal{A}(\mathfrak{D})/\cong| > 1 &\implies \text{source not identifiable from } \mathfrak{D}. \end{aligned}$$

Selecting one isomorphism class in the third case adds information not contained in $\mathfrak{D}$. A prior distribution, regularizer, simplicity criterion, or learned selector may make such a selection, but each supplies additional data.

Compatible families contain field-defining sources only. Configuration, incident source, query, readout, clock, and temporal history remain unselected.

4.3 Consistency and source identification

Set intersection gives

$$\mathcal{A}(\emptyset) = \mathfrak{S}, \quad \mathcal{A}(\mathfrak{D}_1 \cup \mathfrak{D}_2) = \mathcal{A}(\mathfrak{D}_1) \cap \mathcal{A}(\mathfrak{D}_2).$$

Observation order and repetition therefore leave the compatible family unchanged, while additional constraints can only reduce it.

Theorem 4.1 (Source identification criterion). *Let $R_0 \in \mathfrak{S}$ satisfy every constraint in $\mathfrak{D}$. Then*

$$\mathcal{A}(\mathfrak{D})/\cong = \{[R_0]\}$$

if and only if every $R \in \mathfrak{S}$ with $R \not\cong R_0$ violates at least one constraint.

$$\forall R \not\cong R_0, \quad \exists o \in \mathfrak{D} \quad \text{such that} \quad h(R, \iota_o) \notin C_o.$$

Proof. Membership of R_0 makes the compatible family nonempty. Exclusion of every nonisomorphic R leaves only $[R_0]$ in the quotient. Conversely, singleton compatibility excludes every source outside $[R_0]$, so each such source must violate at least one constraint in the defining intersection. $\square$

Exact singleton identification is an algebraic statement. Robust identification under perturbed observations additionally requires topologies on $\mathfrak{S}$ and $\mathcal{Y}$ together with a quantitative separation or continuity estimate.

4.4 Invariance under source isomorphism

Source identification must not depend on coordinate labels. For every source isomorphism $\Phi : R \rightarrow R'$, let the bijections

$$T_{\Phi}^{\mathcal{U}} : \mathcal{U} \longrightarrow \mathcal{U}, \quad T_{\Phi}^{\mathcal{Y}} : \mathcal{Y} \longrightarrow \mathcal{Y}$$

denote the declared transports of interventions and observations. Naturality of h means

$$h(R', T_{\Phi}^{\mathcal{U}} \iota) = T_{\Phi}^{\mathcal{Y}} h(R, \iota).$$

Transported constraint is

$$\Phi o = (T_{\Phi}^{\mathcal{U}} \iota_o, T_{\Phi}^{\mathcal{Y}} C_o).$$

Proposition 4.2 (Isomorphism invariance of compatibility). *If h is natural under source isomorphisms, then*

$$R \in \mathcal{C}(o) \iff R' \in \mathcal{C}(\Phi o).$$

Consequently, an isomorphism-invariant observation collection identifies only source isomorphism classes.

Proof. Naturality gives

$$h(R', T_{\Phi}^{\mathcal{U}} \iota_o) = T_{\Phi}^{\mathcal{Y}} h(R, \iota_o).$$

Invertibility of $T_{\Phi}^{\mathcal{Y}}$ makes membership in C_o equivalent to membership in $T_{\Phi}^{\mathcal{Y}} C_o$. $\square$

4.5 Non-identifiability without role assignments

Untyped records do not determine which coordinates are quantities and which are relations.

Proposition 4.3 (Untyped-record non-identifiability). *No uninterpreted finite record universally determines a localized relation source.*

Proof. Consider three unlabeled cells with one untyped incidence record. Equip both interpretations below with Euclidean carrier metrics, discrete localization algebra $\mathbb{R}^3$, and all coordinate subsets as regions.

One interpretation assigns two cells to the quantity carrier and one to the relation carrier,

$$H_1 = \mathbb{R}^2, \quad G_1 = \mathbb{R}, \quad D_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Another assigns one cell to the quantity carrier and two to the relation carrier,

$$H_2 = \mathbb{R}, \quad G_2 = \mathbb{R}^2, \quad D_2 = \begin{pmatrix} 1 & 1 \end{pmatrix}.$$

Both sources preserve the same three marks and untyped incidence. No source isomorphism relates them because their quantity and relation carrier dimensions differ. Any observation map returning only the untyped record therefore admits both isomorphism classes. $\square$

Non-identifiability occurs before field response. Once observations determine a source up to isomorphism, every representative yields isometrically equivalent derived geometry. [Section 5](#) derives that geometry from D .

5 Relational field geometry

5.1 Active and silent sectors

Fix a localized relation source

$$R = (A, H, G, D).$$

Relation operator D determines two closed quantity sectors,

$$B_R = \overline{\text{ran } D}, \quad N_R = \ker D^*.$$

Active sector B_R contains every quantity variation supported by admitted relations, including limits of such variations. Silent sector N_R contains every quantity direction annihilated by all primitive relation observables.

Theorem 5.1 (Active–silent decomposition). *Quantity carrier decomposes orthogonally as*

$$H = B_R \oplus N_R.$$

Proof. Adjoint definition gives

$$n \in \ker D^* \iff \langle D\eta, n \rangle_H = 0 \quad \text{for every } \eta \in \text{Dom}(D).$$

Thus

$$\ker D^* = (\text{ran } D)^\perp = (\overline{\text{ran } D})^\perp = B_R^\perp.$$

Orthogonal decomposition by a closed subspace yields $H = B_R \oplus B_R^\perp$. □

No closed-range assumption enters this result. Closure in the definition of B_R is essential for general Hilbert sources.

5.2 Orthogonal projection onto the active sector

Projection theorem gives a unique bounded operator

$$P_R : H \longrightarrow H$$

such that $P_R s \in B_R$ and $s - P_R s \in N_R$ for every $s \in H$.

Theorem 5.2 (Active-sector projector). *Orthogonal projector P_R satisfies*

$$P_R^* = P_R, \quad P_R^2 = P_R, \quad \text{ran } P_R = B_R, \quad \ker P_R = N_R, \quad \|P_R\| \leq 1.$$

Any bounded operator with these properties equals P_R .

Proof. Orthogonal projection onto a closed subspace is self-adjoint, idempotent, and contractive. Range and kernel follow from [theorem 5.1](#). Any second operator fixing B_R and annihilating N_R acts identically on both summands of $H = B_R \oplus N_R$. □

Source isomorphisms transport the projector without choosing a basis.

Proposition 5.3 (Naturality of the active projector). *For $\Phi = (\alpha, \phi_\Lambda, U_H, U_G) : R \rightarrow R'$,*

$$U_H B_R = B_{R'}, \quad U_H N_R = N_{R'}, \quad P_{R'} U_H = U_H P_R.$$

Proof. Intertwining identity $U_H D = D' U_G$ transports $\text{ran } D$ onto $\text{ran } D'$. Isometry U_H preserves closure and orthogonality. Uniqueness in [theorem 5.2](#) then gives the projector identity. $\square$

Nonnegative operator DD^* has the same active support,

$$\ker(DD^*) = \ker D^* = N_R,$$

and spectral calculus gives

$$P_R = \mathbf{1}_{(0, \infty)}(DD^*).$$

This identity will connect instantaneous attention to temporal response in Section 11.

5.3 Interpretation of silent directions

A nonzero $n \in N_R$ obeys

$$D^* n = 0, \quad P_R n = 0.$$

Silence means that admitted primitive relations do not detect that quantity direction. It does not mean that the direction is absent from H , that its amplitude is zero, or that a declared readout must annihilate it.

Gauge interpretation requires a separately specified group action whose orbits identify configurations. No such action follows from $D^* n = 0$. Temporal completion may also retain silent initial position and velocity. Section 9 shows that an unforced silent component can undergo affine motion.

Relation-side nullity has a different type. An incident $\eta \in \ker D \subseteq G$ induces no quantity-side source because $D\eta = 0$. Such a direction records relation redundancy rather than a silent quantity configuration. Section 10 compares both null spaces through the polar decomposition and index.

5.4 Global geometry from local relations

Every finite primitive relation contributes one local equation to the silent sector. Using the finite realization of Section 3.4,

$$(D^* n)_r = \frac{1}{w_r} \sum_{p \in X} \mu_p b_{pr} n_p = 0, \quad r \in E.$$

Only sites in the support of b_r enter this equation. Shared sites couple different equations because one amplitude n_p must satisfy all relations incident to p .

Theorem 5.4 (Local characterization of the silent sector). *Let $\{U_i\}$ be a finite regional cover such that, for every primitive relation r , at least one region contains its complete support,*

$$\{r\} \cup \text{supp } b_r \subseteq U_i.$$

Compatible local quantity assignments whose restriction to such a region satisfies $(D^ n)_r = 0$ for every r glue to a unique element of N_R . Conversely, every $n \in N_R$ restricts to compatible local assignments satisfying all primitive equations supported in each region.*

Proof. [Proposition 3.3](#) glues the compatible site values. Complete containment of each relation support places every term of its finite sum in at least one region. Compatibility makes the sum independent of the selected region. Every coordinate of D^*n therefore vanishes, so the glued assignment lies in N_R . Restricting the coordinate identities for any $n \in N_R$ proves the converse. $\square$

Global silent sector is thus the simultaneous solution space of local homogeneous equations. Active sector is its orthogonal complement, and $P_R = I_H - P_{N_R}$ is the unique global projection determined by their closure. No primitive all-to-all interaction is assumed.

An incident source subsequently probes this source-determined geometry. Re-solving the local homogeneous equations for every incident is unnecessary because P_R depends on D , not on the incident.

5.5 Dependence on the source operator

For fixed R , map $s \mapsto P_R s$ is linear. Dependence of P_R on the source operator is not.

Proposition 5.5 (Discontinuity at rank loss). *Let $D \neq 0$ be bounded, define $D_t = tD$, and write $R_t = (A, H, G, D_t)$. Then*

$$P_{R_t} = P_R \quad \text{for } t \neq 0, \quad P_{R_0} = 0.$$

Proof. Nonzero scalar multiplication leaves $\overline{\text{ran } D}$ unchanged. At $t = 0$, the range is $\{0\}$. $\square$

Rank-changing source limits can therefore change the projector discontinuously. Within a norm-continuous fixed-rank finite family, or a closed-range Hilbert family with a uniform spectral gap at zero, spectral projection varies continuously. Section 13 treats its transport.

Deleting one relation may leave B_R unchanged when its section was redundant, or may enlarge N_R when the deleted section carried an independent direction. Restricting $P_R s$ to one region therefore need not equal projection under the truncated regional source.

Source data determine the orthogonal geometry $H = B_R \oplus N_R$ and its regulator P_R . [Section 6](#) uses this geometry to define relational field strength, intrinsic energy, and static variational response.

6 Field strength, energy, and static response

6.1 Configurations and relational field strength

A field configuration is an element

$$a \in \text{Dom}(D^*) \subseteq H.$$

Relational field strength is

$$F_R(a) = D^* a \in G.$$

Each component records the variation detected by one primitive relation direction. Finite weighted incidence gives

$$(F_R(a))_r = \frac{1}{w_r} \sum_{p \in \mathbf{X}} \mu_p b_{pr} a_p.$$

No vanishing condition is imposed on $F_R(a)$. Nonzero field strength is a field value, not automatically a violated constraint.

Field strength is natural under source isomorphisms:

$$\ker F_R = N_R, \quad F_{R'}(U_H a) = U_G F_R(a)$$

for every $\Phi = (\alpha, \phi_\Lambda, U_H, U_G) : R \rightarrow R'$. Its kernel is the silent sector, and covariance follows by taking adjoints of $U_H D = D' U_G$.

6.2 Rigidity of the intrinsic energy

Consider energies obtained by pulling back a continuous quadratic functional $\Phi_G : G \rightarrow \mathbb{R}$ along F_R . Minimal first-order dependence permits no further use of D , D^* , or localization after $F_R(a)$ has been formed. Coordinate independence requires

$$\Phi_G(Ug) = \Phi_G(g)$$

for every orthogonal $U : G \rightarrow G$. Assume also $\Phi_G(0) = 0$ and strict positivity for $g \neq 0$.

Theorem 6.1 (Rigidity of the intrinsic energy). *Every intrinsic energy in this minimal class has the form*

$$V_R(a; 0) = \frac{c}{2} \|F_R(a)\|_G^2 = \frac{c}{2} \|D^* a\|_G^2, \quad c > 0.$$

Scalar c is the remaining positive calibration.

Proof. A continuous quadratic functional on G has a bounded self-adjoint representative T_G ,

$$\Phi_G(g) = \frac{1}{2} \langle g, T_G g \rangle_G.$$

Orthogonal invariance implies $U^* T_G U = T_G$ for every orthogonal U on G . Commutation with reflections makes every one-dimensional subspace invariant; linearity on two-dimensional spans then forces a common eigenvalue. Hence $T_G = c I_G$. Strict positivity gives $c > 0$, and substitution $g = F_R(a)$ proves the claim. $\square$

Uniqueness is conditional on this minimal first-order class. A preferred positive operator Γ on G , inequivalent relation types, higher-order uses of D , or nonlinear dependence on $F_R(a)$ introduces additional constitutive data.

6.3 External-source coupling

Let $j \in H^*$ be a continuous external-source covector with Riesz representative $j^\sharp \in H$. Restriction of j to B_R has Riesz representative $P_R j^\sharp$, since

$$j(b) = \langle j^\sharp, b \rangle_H = \langle P_R j^\sharp, b \rangle_H, \quad b \in B_R.$$

Source-coupled static energy is

$$V_R(a; j) = \frac{c}{2} \|D^* a\|_G^2 - j(P_R a).$$

Projection preserves the form domain and field strength,

$$P_R \text{Dom}(D^*) \subseteq \text{Dom}(D^*), \quad D^* P_R a = D^* a,$$

because $(I_H - P_R)a \in N_R = \ker D^*$. Consequently, only the restriction of j to B_R couples to relational variation. Silent component $(I_H - P_R)j^\sharp$ remains a present ambient covector but produces no intrinsic forcing.

A relation incident $\eta \in \text{Dom}(D)$ induces $j_\eta^\sharp = D\eta \in B_R$. Its coupling has the equivalent intrinsic form

$$j_\eta(P_R a) = \langle D\eta, a \rangle_H = \langle \eta, D^* a \rangle_G.$$

No separate projection is needed after induction because $P_R D\eta = D\eta$.

6.4 Static variational response

For $a, \xi \in \text{Dom}(D^*)$, first variation is

$$\delta V_R(a; j)[\xi] = c \langle D^* a, D^* \xi \rangle_G - j(P_R \xi).$$

This expression defines the weak static response on the form domain. Associated closed quadratic form is

$$\mathfrak{q}_R(a, \xi) = c \langle D^* a, D^* \xi \rangle_G, \quad a, \xi \in \text{Dom}(D^*).$$

Its nonnegative self-adjoint operator is

$$L_R = cDD^*, \quad \text{Dom}(L_R) = \{a \in \text{Dom}(D^*) : D^* a \in \text{Dom}(D)\}.$$

For $a \in \text{Dom}(L_R)$, adjointness and Riesz duality give

$$\delta V_R(a; j)[\xi] = \left\langle L_R a - P_R j^\sharp, \xi \right\rangle_H.$$

Theorem 6.2 (Static variational response). *Strong variational response is*

$$Y_R(a; j) = L_R a - P_R j^\sharp = cDD^* a - P_R j^\sharp.$$

Proof. Expanding $V_R(a + \varepsilon \xi; j)$, subtracting $V_R(a; j)$, dividing by ε , and taking $\varepsilon \rightarrow 0$ gives the weak expression. Membership in $\text{Dom}(L_R)$ permits

$$c \langle D^* a, D^* \xi \rangle_G = \langle cDD^* a, \xi \rangle_H.$$

Self-adjointness of P_R gives $j(P_R \xi) = \langle P_R j^\sharp, \xi \rangle_H$. Riesz uniqueness proves the strong identity. $\square$

Generalized force is $-Y_R(a; j)$, while static equilibrium satisfies $Y_R(a; j) = 0$. Neither object specifies temporal development.

6.5 Boundary variation and source symmetries

Boundary statements require a local differential realization. Suppose a region $U \Subset M$ admits a Green identity

$$\langle D^* a, D^* \xi \rangle_{G(U)} = \langle DD^* a, \xi \rangle_{H(U)} + \mathfrak{b}_U(a, \xi),$$

where $\mathfrak{b}_U$ is the boundary pairing determined by traces of a and ξ on ∂U . Local first variation becomes

$$\delta V_{R,U}(a; j)[\xi] = \langle Y_R(a; j), \xi \rangle_{H(U)} + c \mathfrak{b}_U(a, \xi).$$

Vanishing boundary pairing may follow from compact support, fixed boundary values, or a declared natural boundary condition. [Section 14](#) identifies the corresponding energy current and flux for hyperbolic development.

Let J_H and J_G be skew-adjoint infinitesimal generators satisfying

$$J_H D = D J_G$$

on their common invariant domain. Adjoint intertwining gives $D^* J_H = J_G D^*$, and therefore

$$\left. \frac{d}{d\varepsilon} V_R(e^{\varepsilon J_H} a; 0) \right|_{\varepsilon=0} = c \langle D^* a, J_G D^* a \rangle_G = 0.$$

This static Noether identity is generated by a continuous source symmetry. Coupled energy remains invariant only when the external source and boundary data share that symmetry. Discrete relabelings give equivariance but do not by themselves produce a continuous Noether current.

Field strength, energy, and complete static response are fixed by the source and the external covector. [Section 7](#) restricts changes in that covector to admitted queries and identifies the resulting response difference with attention.

7 Attention as query-restricted response

7.1 Observed queries and induced covectors

Fix a source R and an observed query x constructed without access to its desired response. Input x includes the query, its localization, causally available context, and every admitted record available before the response is known. Observation data determine a compatible relation-incident family

$$\mathcal{I}_R(x) \subseteq \text{Dom}(D)$$

without using the desired response, a concealed target, or an output label. Construction of $\mathcal{I}_R(x)$ belongs to source observation rather than to field variation.

Every $\eta \in \mathcal{I}_R(x)$ induces a continuous covector $j_\eta \in H^*$ by

$$j_\eta(a) = \langle D\eta, a \rangle_H = \langle \eta, D^* a \rangle_G, \quad a \in \text{Dom}(D^*),$$

with Riesz representative

$$j_\eta^\sharp = D\eta.$$

Observed query x therefore determines the compatible covector family

$$\mathcal{J}_R(x) = \{j_\eta : \eta \in \mathcal{I}_R(x)\}.$$

Possible outcomes remain distinct.

1. Empty $\mathcal{I}_R(x)$ means that no relation incident is admitted.
2. Nonempty $\mathcal{I}_R(x)$ with one image under D determines one source covector.
3. Several images under D constitute material query ambiguity.

4. A nonzero incident in $\ker D$ records a present relation-coordinate reading whose induced quantity-side covector is zero.

Case four differs from case one. One contains an admitted relation direction annihilated by D ; the other contains no admitted relation incident.

Compatible readings may differ in G without producing different field couplings. Difference $\eta_1 - \eta_2 \in \ker D$ is invisible on the quantity carrier. Difference outside $\ker D$ changes the source covector and is material to the response.

A query covector is faithful when it represents every reading retained by the observation channel. Canonical determination must be independent of the desired response, natural under source isomorphisms, and faithful in this sense. Assigning one constant covector to every query fails faithfulness whenever retained readings induce different covectors.

7.2 Canonical query covectors

Theorem 7.1 (Canonical query covector). *Suppose $\mathcal{I}_R(x) \neq \emptyset$. Observed query x determines one faithful source covector if and only if*

$$D\eta_1 = D\eta_2 \quad \text{for every } \eta_1, \eta_2 \in \mathcal{I}_R(x).$$

Equivalently, for any $\eta_0 \in \mathcal{I}_R(x)$,

$$\mathcal{I}_R(x) \subseteq \eta_0 + \ker D.$$

When this condition holds,

$$j_x^\# = D\eta_0$$

is independent of the representative.

Proof. Pairwise equality of images makes $D\eta_0$ independent of the chosen reading. Riesz representation then defines one continuous covector j_x , and equality with every j_η gives faithfulness. Conversely, faithfulness requires $j_x^\# = D\eta$ for every compatible reading. Every pair therefore has the same image. Subtracting the images gives the affine-null-class condition. $\square$

Proposition 7.2 (Naturality of query determination). *Canonical query-covector determination is natural under source isomorphisms.*

Proof. Let $\Phi = (\alpha, \phi_\Lambda, U_H, U_G) : R \rightarrow R'$ and suppose

$$\mathcal{I}_{R'}(x') = U_G \mathcal{I}_R(x).$$

For any compatible reading,

$$j_{x'}^\# = D' U_G \eta = U_H D \eta = U_H j_x^\#.$$

No spelling, coordinate order, or preferred representative enters this identity. $\square$

7.3 Admissible query covectors

Let

$$\mathcal{Q}_R \subseteq H^*$$

be the class of continuous localized covectors determined as queries by the observation map. Membership requires independence from the desired response and may follow from [theorem 7.1](#) or another specified observation channel. Query designation is not a property of vector coordinates alone because the same covector can serve a different external role under different observation data.

A query $q \in \mathcal{Q}_R$ changes the external source coordinate

$$j \mapsto j + q$$

while A, H, G, D, D^*, P_R , and L_R remain fixed. It neither writes a new relation nor changes the field geometry.

Relation-coordinate incident

$$[\eta] \in \text{Dom}(D)/\ker D$$

induces the intrinsic query representative

$$q^\sharp = D\eta.$$

Finite and bounded cases have $\text{Dom}(D) = G$, reducing the quotient to $G/\ker D$. This quotient is sufficient whenever the observation channel already determines η modulo $\ker D$. Arbitrary raw coordinates require a separate determination result.

7.4 Query-induced response and attention

Define query-induced change in variational response by

$$\Delta_q Y_R(a; j) = Y_R(a; j + q) - Y_R(a; j).$$

Theorem 7.3 (Query-restricted variational response). *For every $q \in \mathcal{Q}_R$,*

$$\Delta_q Y_R(a; j) = -P_R q^\sharp.$$

A bounded readout $O : H \rightarrow Z$ gives

$$\text{Attn}_{R,O}(q) := -O \Delta_q Y_R(a; j) = O P_R q^\sharp.$$

Proof. Substitution into [theorem 6.2](#) gives

$$(L_R a - P_R j^\sharp - P_R q^\sharp) - (L_R a - P_R j^\sharp) = -P_R q^\sharp.$$

Composition with O proves the second identity. □

Response difference is independent of background configuration a and background source j for the quadratic law established in [section 6](#). Its negative is the query-induced change in generalized force, $P_R q^\sharp$. Attention is the declared readout of that force change.

7.5 Projection and contextual response

Source geometry determines which quantity directions can respond jointly. Query covector specifies the external distinction being requested. Carrier metric supplies $q^\sharp$. Orthogonal projection retains precisely the component supported by admitted relations. Readout observes the declared contextual component.

One projection therefore supplies the comparison, exclusion, regulation, and aggregation duties that conventional attention realizes through learned projections, pairwise scores, masks, normalization, and weighted summation. Equality concerns the resulting causal duty, not intermediate neural arrays.

7.6 Causal criteria for attention

Source-preserving interventions identify the response as attention.

1. Changing a query changes the response through $P_R q^\sharp$.
2. Changing an admitted relation may change P_R and the response to a fixed query.
3. Covectors annihilated by P_R cannot affect the intrinsic query response.
4. Causal availability conditions exclude future or masked source components.
5. Source isomorphisms transport responses without changing their structural content.
6. Context-free pooling and output-only controls cannot satisfy all these interventions.

Together these properties identify query-conditioned contextual selection rather than agreement on one output label.

7.7 Query-restricted and complete response

Attention is the query-induced difference $-P_R q^\sharp$ of the static variational response. It is not the complete response $Y_R(a; j)$, the intrinsic operator L_R , or a realized temporal history.

Different sources may have the same P_R and therefore identical instantaneous attention for every admitted query while retaining different DD^* , spectra, and causal responses. [Sections 11](#) and [15](#) give an exact finite witness.

[Section 8](#) treats changes to the admitted relations themselves. Such source variation redetermines P_R and L_R , rather than merely probing the fixed geometry through $j \mapsto j + q$.

8 Source variation and geometric redetermination

8.1 Configuration and source variations

A source variation is an admissible map between localized relation sources,

$$\sigma : R \mapsto R'.$$

It may change the localization algebra, regional supports, either carrier, either carrier metric, or the relation operator. Derived geometry must then be determined from R' ,

$$B_{R'} = \overline{\text{ran } D'}, \quad N_{R'} = \ker D'^*, \quad P_{R'} = P_{B_{R'}}, \quad L_{R'} = cD'D'^*.$$

No equality between old and new objects is assumed. Comparing them directly requires common carriers or a declared transport map. Finite identities below hold on one fixed quantity carrier with one fixed metric.

Source variation differs from configuration variation. Replacing a by $a + \varepsilon \xi$ moves within the form domain of one source. Replacing R by R' changes the geometry, energy, or response law governing subsequent configurations.

8.2 Finite relation updates

Fix a finite source R and hold its quantity carrier, metric, and existing relation sections fixed. Let $\mathcal{C}_{\text{add}} \subseteq H$ be the span of finitely many newly admitted relation sections. New active sector is

$$B_{R_{\text{add}}} = B_R + \mathcal{C}_{\text{add}}.$$

Only the residual subspace

$$\mathcal{R}_{\text{add}} = (I_H - P_R)\mathcal{C}_{\text{add}}$$

adds active directions.

Theorem 8.1 (Finite relation addition). *Finite relation addition satisfies*

$$B_{R_{\text{add}}} = B_R \oplus \mathcal{R}_{\text{add}}, \quad P_{R_{\text{add}}} = P_R + P_{\mathcal{R}_{\text{add}}}.$$

Proof. Residual subspace lies in $B_R^\perp$. Every $c \in \mathcal{C}_{\text{add}}$ decomposes as $c = P_R c + (I_H - P_R)c$, so $B_R + \mathcal{C}_{\text{add}} = B_R + \mathcal{R}_{\text{add}}$. Orthogonality makes this sum direct and makes the projectors additive. $\square$

For one added section b , define

$$r_b = (I_H - P_R)b.$$

If $r_b \neq 0$, then

$$P_{R_{\text{add}}} s = P_R s + r_b \frac{\langle r_b, s \rangle_H}{\langle r_b, r_b \rangle_H}.$$

If $r_b = 0$, the addition changes the source without changing its active projector. Intrinsic operator $D'D'^*$ may still change.

Removal requires the complete surviving source. Let $B_{R_{\text{rem}}} \subseteq B_R$ be the span generated by all relations that remain and define

$$\mathcal{L}_{\text{rem}} = B_R \cap B_{R_{\text{rem}}}^\perp.$$

Theorem 8.2 (Finite relation removal). *Finite relation removal satisfies*

$$B_R = B_{R_{\text{rem}}} \oplus \mathcal{L}_{\text{rem}}, \quad P_{R_{\text{rem}}} = P_R - P_{\mathcal{L}_{\text{rem}}}.$$

Proof. Inclusion $B_{R_{\text{rem}}} \subseteq B_R$ gives the displayed orthogonal decomposition. Subtracting the projector onto the lost summand leaves the projector onto the surviving summand. $\square$

Subtracting the rank-one projector of a deleted section is generally invalid because another surviving relation may generate the same direction. Metric changes, carrier changes, and unbounded-operator changes also fall outside these fixed-carrier identities.

8.3 Composition of source updates

Repeated additions and removals produce several recognizable effects.

Source change	Derived geometric effect
Independent relation addition	Active rank increases
Redundant relation addition	P_R is unchanged, although D and L_R may change
Relation joining formerly independent active subspaces	Previously separated response directions become coupled
Removal of the last independent connecting directions	Active geometry separates and rank may decrease
Replacement of one relation by another	Addition and removal identities compose
Change of carrier-metric ratios	Adjoint, orthogonality, and projector may change at fixed incidence
Change of relation weights	Intrinsic response may change while P_R remains fixed

Names such as write, revise, merge, split, reweight, and delete describe these derived transition signatures only after an observation criterion assigns the name. None is an additional primitive field operation.

Final geometry depends on the final admitted source, not on an arbitrary order used to evaluate exact updates. Path dependence can enter only when the admission rule, metrics, coefficients, or realized temporal state retain history.

Equal projectors do not imply equal sources. Cycle-closing additions and redundant removals can preserve P_R while changing D , D^* , L_R , field strength, or causal response. A zero change in attention therefore does not prove absence of structural change.

8.4 Queries under source variation

A query is a fixed-source perturbation,

$$(R, j) \mapsto (R, j + q).$$

A source variation changes the field-defining data,

$$(R, j) \mapsto (R', j'),$$

where transport of j must be declared if the carrier changes. For fixed common carrier, metric, query covector, and readout,

$$\text{Attn}_{R',O}(q) - \text{Attn}_{R,O}(q) = O(P_{R'} - P_R)q^\sharp.$$

This identity measures the query-visible part of source variation. It does not detect changes confined to $D'D'^* - DD^*$, relation-side nullity, or readout-invisible sectors.

Querying therefore evaluates one fixed-source response law. Source variation redetermines the law used by subsequent queries. [Section 9](#) derives temporal development for a fixed source, while [section 13](#) treats smooth and rank-changing source families.

9 Intrinsic temporal dynamics

9.1 Static energy does not determine temporal dynamics

Static response does not by itself determine temporal development. For fixed source R , both

$$\dot{a} + L_R a = P_R j^\sharp$$

and

$$m\ddot{a} + L_R a = P_R j^\sharp$$

use the same intrinsic operator and have the same stationary equation. Their histories differ. In the source-free case, an eigenvector of L_R with eigenvalue $\lambda > 0$ decays as $e^{-\lambda\tau}$ under the first law and oscillates with angular frequency $\sqrt{\lambda/m}$ under the second.

Proposition 9.1 (Static data do not select a temporal law). *A localized relation source and its static energy do not select a temporal law.*

Proof. Both displayed laws are local in the clock coordinate, source-natural, and compatible with the same static equilibrium. Distinct histories on any positive spectral mode prevent their identification. $\square$

Dissipative development additionally supplies a force-to-rate relation and energy loss, whereas inertial development retains field energy. Static data alone cannot choose between these temporal classes.

9.2 Metric clock and inertial calibration

A metric clock is a one-dimensional affine interval I_τ with metric $d\tau^2$, orientation, and rate calibration. It orders histories and measures their rates. It neither supplies an initial state nor selects a temporal equation by itself.

A history is a map

$$a : I_\tau \longrightarrow H$$

in the local energy class

$$a \in W_{\text{loc}}^{1,2}(I_\tau; H) \cap L_{\text{loc}}^2(I_\tau; \text{Dom}(D^*)).$$

Clock velocity $\dot{a} \in H$ and relational field strength $D^*a \in G$ have different types. Their complete first-jet carrier is the orthogonal sum

$$\mathbb{J}_R = H \oplus G, \quad \widehat{F}_R(a) = (\dot{a}, D^*a).$$

Source isomorphism $\Phi = (\alpha, \phi_\Lambda, U_H, U_G) : R \rightarrow R'$ transports this jet by

$$\widehat{F}_{R'}(U_H a) = (U_H \dot{a}, U_G D^*a).$$

No identification of H with G is required. Positive constant m calibrates the inertial norm on the clock derivative.

9.3 Closed reversible variational class

Consider local first-order actions built only from the supplied metric clock, carrier metrics, relation operator, static energy, and external-source coupling. Admissible densities satisfy the following conditions.

1. Configuration dependence occurs through a , $\dot{a}$, D^*a , and $j(\tau)(P_Ra)$, with no higher clock or relational derivatives.
2. Source isomorphisms and orthogonal changes of carrier coordinates preserve the action.
3. Time reversal $\tau \mapsto -\tau$, $\dot{a} \mapsto -\dot{a}$ changes the density by at most a total derivative.
4. Euler–Lagrange equations have a field-independent second-order principal part, and setting $\dot{a} = 0$ recovers the static energy and coupling of [section 6](#).
5. Source-free Legendre energy is nonnegative and positive on every nonzero active field strength or velocity.

Any additional inertia tensor, carrier map, or time-odd coefficient enlarges the supplied data and lies outside this minimal class.

9.4 Rigidity of the action

Theorem 9.2 (Rigidity of the intrinsic temporal action). *Within the closed reversible variational class, every temporal action is equivalent, up to multiplication by a positive constant, addition of a total clock derivative, and a configuration-independent term, to*

$$\mathcal{S}_R[a; j] = \int_{I_\tau} \left[\frac{m}{2} \|\dot{a}\|_H^2 - \frac{c}{2} \|D^*a\|_G^2 + j(\tau)(P_Ra) \right] d\tau, \quad m, c > 0.$$

Consequently, every admitted action has the same second-order wave principal part.

Proof. Fix one clock point and write $v = \dot{a} \in H$ and $g = D^*a \in G$. Metric-only dependence and orthogonal invariance make the velocity density a function $\varphi(\|v\|_H^2/2)$. Its Hessian with respect to v is

$$\varphi'(s)I_H + \varphi''(s)v \otimes v, \quad s = \frac{1}{2}\|v\|_H^2.$$

Field independence of the clock-principal part forces $\varphi'' = 0$, so positivity gives $\varphi(s) = ms + b_H$ with $m > 0$.

Orthogonal invariance on G likewise makes the field-strength density a function $\psi(r)$ of

$$r = \frac{1}{2}\|g\|_G^2.$$

Its Hessian with respect to g is

$$\psi'(r)I_G + \psi''(r)g \otimes g.$$

Field independence of the spatial principal part forces $\psi'' = 0$. Hence $\psi(r) = cr + b_G$, and positivity gives $c > 0$. Static compatibility fixes the potential sign and the linear coupling to $j(P_Ra)$.

A mixed pairing of v and g would require an independently supplied zero-order map between H and G . Using the source operator gives

$$\langle D^*\dot{a}, D^*a \rangle_G = \frac{1}{2} \frac{d}{d\tau} \|D^*a\|_G^2,$$

which is a temporal boundary term for fixed R . Quadratic dependence on D^*a has higher mixed order and violates the declared principal class. Time-reversal invariance excludes non-boundary terms odd in v .

Kinetic-minus-potential sign gives positive Legendre energy. Reversing the potential sign produces $m\ddot{a} - L_R a = 0$ and hyperbolic growth on every positive spectral mode. Remaining freedom consists of the calibrations and variationally neutral terms stated above. $\square$

Uniqueness belongs to the declared closed reversible variational class.

9.5 Information-field equation

Compactly supported history variation $a + \varepsilon\xi$ gives

$$\left. \frac{d}{d\varepsilon} \mathcal{S}_R[a + \varepsilon\xi; j] \right|_{\varepsilon=0} = \int_{I_\tau} \left[m\langle \dot{a}, \dot{\xi} \rangle_H - c\langle D^*a, D^*\xi \rangle_G + j(\tau)(P_R\xi) \right] d\tau.$$

Integration by parts in τ and use of the operator associated with the closed form $c\|D^*a\|_G^2$ yield

$$m\ddot{a} + L_R a = P_R j(\tau)^\sharp, \quad L_R = cDD^*.$$

This abstract wave equation is the uniquely determined temporal law within the closed reversible variational class. Local differential realizations become normally hyperbolic under the principal-symbol hypotheses of [section 14](#). Static response reappears as

$$m\ddot{a} + Y_R(a; j(\tau)) = 0.$$

Active and silent components evolve independently for fixed R . Writing $a = a_B + a_N$ with $a_B = P_R a$ and $a_N = P_{N_R} a$ gives

$$m\ddot{a}_B + L_R a_B = P_R j(\tau)^\sharp, \quad m\ddot{a}_N = 0.$$

Silent configurations carry no intrinsic restoring force. They are not thereby gauge directions and need not vanish.

9.6 Global well-posedness

Set

$$\mathcal{H}_R^{\text{en}} = \text{Dom}(L_R^{1/2}) \oplus H = \text{Dom}(D^*) \oplus H$$

with graph-energy norm

$$\|(a, v)\|_{\mathcal{H}_R^{\text{en}}}^2 = \|a\|_H^2 + \|L_R^{1/2} a\|_H^2 + \|v\|_H^2,$$

and define

$$\Omega_R = (L_R/m)^{1/2}.$$

Theorem 9.3 (Global finite-energy well-posedness). *Every initial pair*

$$a_0 \in \text{Dom}(L_R^{1/2}), \quad v_0 \in H$$

determines a unique global finite-energy weak solution of the source-free equation. Spectral calculus gives

$$a(\tau) = \cos(\tau\Omega_R)a_0 + \Omega_R^{-1} \sin(\tau\Omega_R)v_0,$$

where the second multiplier is interpreted as τI_{N_R} on $\ker L_R = N_R$. For $P_R j^\sharp \in L^1_{\text{loc}}(I_\tau; H)$, Duhamel integration gives the unique forced weak solution

$$a(\tau) = \cos(\tau\Omega_R)a_0 + \Omega_R^{-1} \sin(\tau\Omega_R)v_0 + \frac{1}{m} \int_0^\tau \Omega_R^{-1} \sin((\tau-s)\Omega_R) P_R j(s)^\sharp ds.$$

Proof. Spectral theorem reduces the equation to one scalar oscillator for each positive spectral value and free affine motion at zero. Cosine and sinc multipliers are bounded on $\mathcal{H}_R^{\text{en}}$ over every finite clock interval. Standard approximation by spectral cutoffs proves existence, uniqueness, and continuous dependence for energy data. Bochner integration gives the forced solution under the stated local integrability condition. $\square$

Stronger initial data and forcing give the corresponding strong solution. No compactness or finite-rank assumption is required.

9.7 Conservation of energy

Source-free mechanical energy is

$$E_R(a, \dot{a}) = \frac{m}{2} \|\dot{a}\|_H^2 + \frac{1}{2} \|L_R^{1/2} a\|_H^2 = \frac{m}{2} \|\dot{a}\|_H^2 + \frac{c}{2} \|D^* a\|_G^2.$$

Proposition 9.4 (Energy conservation). *Every source-free finite-energy solution conserves E_R . Mechanical energy is nonnegative but does not control the initial position in N_R , which is why the Cauchy carrier uses the full graph norm. A sufficiently regular forced solution satisfies*

$$\frac{d}{d\tau} E_R(a, \dot{a}) = \langle P_R j(\tau)^\sharp, \dot{a} \rangle_H.$$

If j is independent of τ , coupled energy

$$E_R^j(a, \dot{a}) = E_R(a, \dot{a}) - j(P_R a)$$

is conserved.

Proof. Pairing the field equation with $\dot{a}$ gives the work identity for strong solutions. Spectral approximation extends it to the energy class. Constant forcing converts the right side into $d[j(P_R a)]/d\tau$. $\square$

Positive energy and time reversal distinguish this development from the dissipative alternative used at the beginning of the section. [Section 10](#) expresses the same inertial law as first-order exchange between quantity and relation carriers.

10 First-order quantity–relation dynamics

10.1 Quantity and relation variables

Second-order development has an equivalent first-order representation on the graded carrier $H \oplus G$. Define

$$v = \sqrt{m} \dot{a} \in H, \quad u = \sqrt{c} D^* a \in G.$$

Variable v carries clock-directed quantity motion, while u carries relational field strength. Configuration-generated data obey the compatibility condition

$$u = \sqrt{c} D^* a.$$

Arbitrary vectors in $H \oplus G$ need not satisfy this condition. In particular, a nonzero component of u in $\ker D$ cannot equal $D^* a$, because $\text{ran } D^* \perp \ker D$.

10.2 First-order exchange equation

Define

$$\mathbb{A}_D = \begin{pmatrix} 0 & -D \\ D^* & 0 \end{pmatrix}, \quad \text{Dom}(\mathbb{A}_D) = \text{Dom}(D^*) \oplus \text{Dom}(D).$$

Theorem 10.1 (First-order exchange representation). *Every sufficiently regular solution of the information-field equation determines a solution of*

$$\frac{d}{d\tau} \begin{pmatrix} v \\ u \end{pmatrix} = \sqrt{\frac{c}{m}} \mathbb{A}_D \begin{pmatrix} v \\ u \end{pmatrix} + \begin{pmatrix} P_R j(\tau)^\sharp / \sqrt{m} \\ 0 \end{pmatrix}.$$

Conversely, a solution of this equation reconstructs the second-order field history after choosing a_0 , integrating $\dot{a} = v/\sqrt{m}$, and imposing $u(0) = \sqrt{c} D^ a_0$.*

Proof. Differentiation of the scaled variables and substitution of $m\ddot{a} + cDD^*a = P_R j^\sharp$ give

$$\dot{v} = -\sqrt{\frac{c}{m}} Du + \frac{1}{\sqrt{m}} P_R j^\sharp, \quad \dot{u} = \sqrt{\frac{c}{m}} D^* v.$$

Conversely, integrate $\dot{a} = v/\sqrt{m}$ and define the compatibility defect

$$e = u - \sqrt{c} D^* a.$$

Second block gives $\dot{e} = 0$. Initial compatibility therefore persists, and the first block recovers the second-order field equation. $\square$

10.3 Skew-adjoint generator and energy balance

Theorem 10.2 (Skew-adjoint exchange generator). *Operator $\mathbb{A}_D$ is skew-adjoint on $H \oplus G$.*

Proof. For two vectors in its domain,

$$\begin{aligned} \langle \mathbb{A}_D(v, u), (x, y) \rangle_{H \oplus G} &= -\langle Du, x \rangle_H + \langle D^* v, y \rangle_G \\ &= -\langle u, D^* x \rangle_G + \langle v, Dy \rangle_H \\ &= -\langle (v, u), \mathbb{A}_D(x, y) \rangle_{H \oplus G}. \end{aligned}$$

Thus $\mathbb{A}_D$ is skew-symmetric. If (x, y) belongs to its adjoint domain, boundedness against $u \in \text{Dom}(D)$ and $v \in \text{Dom}(D^*)$ independently forces

$$x \in \text{Dom}(D^*), \quad y \in \text{Dom}(D^{**}) = \text{Dom}(D).$$

Adjoint action on this domain is $(Dy, -D^*x) = -\mathbb{A}_D(x, y)$, proving $\mathbb{A}_D^* = -\mathbb{A}_D$. $\square$

Stone's theorem gives the orthogonal group

$$e^{\tau \sqrt{c/m} \mathbb{A}_D}.$$

First-order energy

$$\mathcal{E}_R^{(1)}(v, u) = \frac{1}{2} \|v\|_H^2 + \frac{1}{2} \|u\|_G^2$$

satisfies

$$\frac{d}{d\tau} \mathcal{E}_R^{(1)}(v, u) = \langle \dot{a}, P_R j(\tau)^\# \rangle_H.$$

On the compatible sector, $\mathcal{E}_R^{(1)} = E_R$. Skew exchange transfers energy between the two carriers without changing their source-free sum.

10.4 Second-order reduction

Block multiplication gives

$$\mathbb{A}_D^2 = - \begin{pmatrix} DD^* & 0 \\ 0 & D^*D \end{pmatrix}$$

on its natural square domain. Source-free first-order variables consequently satisfy

$$\ddot{v} + \frac{c}{m} DD^* v = 0, \quad \ddot{u} + \frac{c}{m} D^* D u = 0.$$

Quantity-side and relation-side waves are the two graded components of one skew-adjoint exchange law. Their nonzero frequencies agree, although their null spaces may have different dimensions.

10.5 Polar decomposition and spectral pairing

Let

$$D = U_D |D|, \quad |D| = (D^* D)^{1/2}, \quad Q_R = P_{\overline{\text{ran } D^*}}.$$

Proposition 10.3 (Polar-decomposition support identities). *Partial isometry U_D satisfies*

$$U_D U_D^* = P_R, \quad U_D^* U_D = Q_R, \quad DD^* = U_D |D|^2 U_D^*$$

on the corresponding supports. Consequently, DD^ and D^*D have identical nonzero spectral multiplicities.*

Proof. Initial space of U_D is $\overline{\text{ran } D^*}$, and its final space is $\overline{\text{ran } D} = B_R$. Standard polar-decomposition identities give the projectors and spectral intertwining. $\square$

Projector P_R records active quantity support, positive operator $|D|$ records modal strength, and partial isometry U_D pairs relation modes with quantity modes. Instantaneous attention retains only P_R , while temporal response also depends on modal strengths.

10.6 Null spaces and index

Quantity null sector

$$\ker D^* = N_R$$

contains configurations with zero relational field strength. Relation null sector $\ker D$ contains primitive-relation combinations producing no quantity-side incidence. Neither sector is automatically gauge.

For finite-dimensional carriers and $r = \text{rank } D$, rank-nullity gives

$$\dim H = r + \dim \ker D^*, \quad \dim G = r + \dim \ker D.$$

Whenever D is Fredholm,

$$\text{ind } D = \dim \ker D - \dim \ker D^*.$$

Index records the imbalance between relation redundancies and silent quantity directions. Fixed-source first-order evolution preserves both null sectors. [Section 11](#) uses the shared positive spectrum to distinguish instantaneous attention from complete causal response.

11 Causal and spectral response

11.1 Retarded response

Zero-past response to a time-dependent source is determined by the retarded kernel

$$K_R(\tau) = \mathbf{1}_{[0,\infty)}(\tau) \frac{1}{m} \Omega_R^{-1} \sin(\tau \Omega_R) P_R.$$

Spectral multiplier $\Omega_R^{-1} \sin(\tau \Omega_R)$ is interpreted as τ at zero frequency. Right multiplication by P_R removes the silent source component before propagation.

For a past-integrable source history with vanishing past data,

$$a(\tau) = \int_{-\infty}^{\tau} K_R(\tau - s) j(s)^\sharp ds.$$

Retarded support enforces causality. A source applied after clock value τ_0 cannot change the field at any earlier value.

11.2 Response jets and the intrinsic operator

Bounded L_R permits the norm-convergent expansion

$$K_R(\tau) = \mathbf{1}_{[0,\infty)}(\tau) \sum_{k=0}^{\infty} \frac{(-1)^k \tau^{2k+1}}{(2k+1)! m^{k+1}} L_R^k P_R.$$

Proposition 11.1 (Odd response jets). *If L_R is bounded, its operator-norm right response jets satisfy*

$$K_R^{(2k+1)}(0^+) = \frac{(-1)^k}{m^{k+1}} L_R^k P_R, \quad k \geq 0,$$

while even right jets vanish. If L_R is unbounded, the same identities hold strongly on $\text{Dom}(L_R^k)$. In particular,

$$K'_R(0^+) = \frac{1}{m}P_R, \quad K'''_R(0^+) = -\frac{1}{m^2}L_R.$$

Proof. Spectral expansion of the sine gives the norm-convergent series and its right derivatives. Spectral truncation gives the strong identities on the stated domains in the unbounded case. $\square$

First nonzero jet recovers active support after calibration. Next nonzero jet recovers intrinsic stiffness. Later jets recover successive powers of the same operator and therefore reveal its higher-order propagation structure.

11.3 Instantaneous attention and causal response

Instantaneous attention depends on P_R . Causal response depends on the complete spectral measure of L_R visible through the incident and readout ports.

Theorem 11.2 (Attention and causal-response separation). *Suppose two finite sources on the same H satisfy $P_{R_1} = P_{R_2} = P$. Let q be an admitted query and assume, for some $k \geq 1$,*

$$L_{R_1}^\ell Pq^\sharp = L_{R_2}^\ell Pq^\sharp \quad (0 \leq \ell < k), \quad L_{R_1}^k Pq^\sharp \neq L_{R_2}^k Pq^\sharp.$$

Every instantaneous attention response agrees, response jets through order $2k - 1$ agree for q , and jets of order $2k + 1$ differ. Hence the two retarded responses differ.

Proof. Equality of projectors gives identical attention for every query. [Proposition 11.1](#) gives equality of earlier jets and inequality of the stated jet. Distinct derivatives preclude equality of the finite-dimensional analytic kernels. $\square$

Attention therefore observes the support term of a larger causal law. It cannot determine spectral stiffness from instantaneous responses alone.

11.4 Frequency-domain response

Use Fourier convention

$$\widehat{a}(\omega) = \int_{\mathbb{R}} e^{i\omega\tau} a(\tau) d\tau.$$

Retarded susceptibility is the boundary value

$$\chi_R(\omega) = \lim_{\epsilon \downarrow 0} P_R [L_R - m(\omega + i\epsilon)^2 I_H]^{-1} P_R$$

in the weak or distributional sense. Away from the spectrum of Ω_R , it obeys

$$\widehat{a}(\omega) = \chi_R(\omega) \widehat{j}^\sharp(\omega).$$

Poles or spectral singularities occur at intrinsic frequencies. Residues record the corresponding port-visible spectral projectors. Static susceptibility is the zero-frequency inverse only on the positive spectral sector and only when zero is spectrally separated.

11.5 Reciprocity

Proposition 11.3 (Reciprocity of the retarded kernel). *For every $\tau > 0$, retarded kernel $K_R(\tau)$ is self-adjoint. Consequently,*

$$\langle x, K_R(\tau)y \rangle_H = \langle K_R(\tau)x, y \rangle_H.$$

Proof. Kernel is a real Borel function of self-adjoint L_R , composed with its commuting spectral support P_R . $\square$

Self-adjointness therefore makes the intrinsic closed response reciprocal.

11.6 Reachability, observability, and minimal realization

Assume finite-dimensional H , an admitted incident class $\mathcal{J}_0 \subseteq H^*$, and readout $O : H \rightarrow Z$. Define

$$\mathcal{R}_{R,\mathcal{J}_0} = \text{span}\{L_R^k P_R j^\# \mid k \geq 0, j \in \mathcal{J}_0\}$$

and

$$\mathcal{N}_{R,O} = \bigcap_{k \geq 0} \ker(OL_R^k).$$

First subspace is reachable from the specified incidents. Second is invisible to every readout response jet. Cayley–Hamilton reduces both definitions to powers below $\dim H$.

Only

$$\mathcal{R}_{R,\mathcal{J}_0} / (\mathcal{R}_{R,\mathcal{J}_0} \cap \mathcal{N}_{R,O})$$

can be identified from the declared transfer behavior. Hidden invariant sectors may vary without changing any observed input–output response.

11.7 Identifiability from response data

Complete retarded response determines the spectral measure of L_R only on the reachable and observable sector. In finite dimensions, unique decomposition of the response into modal sine functions recovers visible frequencies and their port weights. Response jets provide the equivalent moment sequence $OL_R^k P_R j^\#$.

Theorem 11.4 (Transfer-equivalent realizations are similar). *Two finite-dimensional controllable and observable first-order realizations with identical transfer functions are similar. Equality of transfer behavior does not, by itself, identify their Hilbert metrics, quantity–relation gradings, or relation operators.*

Proof. Equal transfer functions give equal Markov parameters. Map each spanning vector generated by one state operator and incident port to the corresponding vector in the other realization. Observability makes this map well-defined and injective, while controllability makes it surjective. It intertwines state, incident, and readout maps and is therefore a similarity. General similarity need not be orthogonal or factor as carrier isometries obeying $U_H D = D' U_G$. $\square$

Even complete knowledge of $L_R = cDD^*$ leaves right-orthogonal freedom in D and may leave invisible relation-null directions. Recovering a localized relation source requires the stronger typed observation conditions of Section 4.

11.8 Spectral response at fixed ports

Assume finite-dimensional H , let $B_{\text{in}} : U \rightarrow B_R \subseteq H$ be an active incident port, and write the spectral resolution of the intrinsic operator as

$$L_R = \sum_{\lambda \geq 0} \lambda E_\lambda.$$

Fixed incident and observation ports determine the spectral residues

$$W_\lambda = O E_\lambda B_{\text{in}}.$$

Every causal port response is determined by these residues because

$$O f(L_R) B_{\text{in}} = \sum_{\lambda} f(\lambda) W_\lambda$$

for every function f defined on the finite spectrum.

Write

$$\mathcal{R}_{R,B_{\text{in}}} = \text{span}\{L_R^k B_{\text{in}} u : k \geq 0, u \in U\}$$

and represent its reachable-and-observable quotient by

$$\mathcal{K}_{R,B_{\text{in}},O} = \mathcal{R}_{R,B_{\text{in}}} \ominus (\mathcal{R}_{R,B_{\text{in}}} \cap \mathcal{N}_{R,O}).$$

Proposition 11.5 (Port-visible spectral dimension). *Reachable and observable dimension in the λ -eigenspace equals $\text{rank } W_\lambda$. Hence*

$$\dim \mathcal{K}_{R,B_{\text{in}},O} = \sum_{\lambda > 0} \text{rank } W_\lambda.$$

Proof. Powers of L_R act by scalar multiplication in one eigenspace. Reachable directions there are $\text{ran}(E_\lambda B_{\text{in}})$. Quotient by their intersection with $\ker O$ has dimension $\text{rank}(O E_\lambda B_{\text{in}})$. Since $\text{ran } B_{\text{in}} \subseteq B_R$, the zero eigenspace contributes nothing. Orthogonality of positive spectral subspaces makes the dimensions additive. $\square$

Force-to-position transfer function on this carrier is

$$H(s) = \sum_{\lambda > 0} \frac{W_\lambda}{ms^2 + \lambda}.$$

Distinct pole pairs and their residue ranks give continuous-time first-order McMillan degree

$$2 \sum_{\lambda > 0} \text{rank } W_\lambda.$$

Sampling can merge frequencies or suppress a port at a particular step. Continuous-time minimality therefore does not imply minimality of every sampled realization.

11.9 Source symmetries and conserved sectors

Let a source automorphism be represented by orthogonal carrier maps U_H and U_G satisfying

$$U_H D = D U_G.$$

Such an automorphism commutes with intrinsic response because

$$U_H L_R U_H^* = c U_H D D^* U_H^* = c D U_G U_G^* D^* = L_R.$$

Assume additionally that U_H is involutive and set

$$\Pi_{\pm} = \frac{1}{2}(I_H \pm U_H).$$

Proposition 11.6 (Parity-sector decomposition). *Suppose ranges of the incident and specified initial-state ports are invariant under U_H , and suppose $\text{ran } O^*$ is invariant under U_H . Reachable and unobservable carriers are then invariant under $\Pi_{\pm}$, and the minimal response carrier decomposes as their orthogonal parity sectors. Every causal response at these ports equals the sum of its two sector responses.*

Proof. Commutation with L_R carries port invariance through every Krylov power, so reachable carrier is invariant. Invariance of $\text{ran } O^*$ is equivalent to invariance of $\ker O$. Commutation then preserves every $\ker(OL_R^k)$, their intersection, and its intersection with the reachable carrier. Orthogonal complementation inside the reachable carrier preserves the same decomposition. Functional calculus of L_R acts independently on both sectors, proving response equality. $\square$

Symmetry of L_R alone does not imply this commutation with minimal reduction. A noninvariant incident-port map can inject coordinates into both parity sectors, while a degenerate eigenspace can remain fully reachable and observable.

Continuous source symmetries give a separate conservation statement. Let $J^* = -J$ and $[J, L_R] = 0$. Every source-free solution conserves

$$Q_J(a, \dot{a}) = m \langle \dot{a}, Ja \rangle_H.$$

Under forcing f , its derivative is $\langle f, Ja \rangle_H$. A conserved sector may therefore remain observable and cannot be removed merely because it is conserved.

11.10 Null directions and declared gauge tangents

Quantity and relation null spaces have different operational effects. Directions in $\ker D^*$ have zero intrinsic stiffness, while directions in $\ker D$ produce zero quantity-side incident. An initial-state port and readout may still expose motion in $\ker D^*$. Relation-null directions instead disappear only from response to their corresponding relation coordinates.

No gauge quotient follows from either null space alone. A declared linear gauge tangent carrier $\mathcal{G}_{\text{gauge}} \subset H$ admits the present quotient only when

$$\mathcal{G}_{\text{gauge}} \subseteq \ker D^*, \quad O \mathcal{G}_{\text{gauge}} = 0, \quad \mathcal{G}_{\text{gauge}} \perp \text{ran } B_{\text{in}}.$$

These conditions distinguish a declared redundant action from a silent but physical zero-frequency mode. Reachable-and-observable reduction removes a compatible gauge tangent without deleting an observable silent mode.

11.11 Rank and state-dimension lower bounds

Let $\mathcal{L} : U \rightarrow Z$ be a fixed finite response map. Any exact linear factorization $\mathcal{L} = AB$ through a carrier W satisfies

$$\dim W \geq \text{rank } \mathcal{L}.$$

Singular-value factorization attains equality. Rank is therefore the exact intermediate-dimension bound among linear factorizations, but not an arithmetic-operation lower bound for unrestricted algorithms.

Coordinate access has a different exact statement. If inputs range over all of U in a fixed coordinate basis, every nonzero column of $\mathcal{L}$ must be inspected in the worst case. Inputs differing only in that coordinate otherwise give the same computation and different required outputs. Likewise, every nonzero output row must be produced when the output is explicitly materialized.

These bounds apply to any finite static or fixed-time response map. A basis incident can select one stored column, so rank alone supplies no positive arithmetic lower bound before output materialization.

Continuous-time response has the McMillan-degree bound derived above. Sampled response instead has minimal discrete linear state dimension equal to the stabilized rank of the block Hankel matrix

$$\mathcal{H}_{A,B,C} = [CA^{i+j}B]_{i,j \geq 0}.$$

Finite state dimension n requires at most n block rows and columns by Cayley–Hamilton. Stabilized Hankel rank detects degree loss caused by sampling aliases or suppressed ports.

[Section 12](#) realizes the same causal law on the doubled quantity state $(a, \dot{a})$ and states the exact relation to stateful attention.

12 State-space realization and stateful attention

12.1 State-space representation

Second-order field development can be written on the doubled quantity carrier. Let

$$\mathbf{x} = \begin{pmatrix} a \\ \dot{a} \end{pmatrix}, \quad \mathcal{G}_R = \begin{pmatrix} 0 & I_H \\ -m^{-1}L_R & 0 \end{pmatrix}.$$

Field equation becomes

$$\dot{\mathbf{x}} = \mathcal{G}_R \mathbf{x} + \begin{pmatrix} 0 \\ m^{-1}P_R j^\# \end{pmatrix}.$$

Energy carrier is $\mathcal{H}_R^{\text{en}}$, and the generator domain is

$$\text{Dom}(\mathcal{G}_R) = \text{Dom}(L_R) \oplus \text{Dom}(L_R^{1/2}).$$

This representation retains only quantity configuration and velocity. [Section 10](#) instead exposes the underlying quantity–relation exchange on $H \oplus G$.

An impulsive query $q\delta_{\tau_0}$ gives the junction

$$\mathbf{x}(\tau_0^+) - \mathbf{x}(\tau_0^-) = \begin{pmatrix} 0 \\ m^{-1}P_R q^\# \end{pmatrix}.$$

Configuration remains continuous, while canonical momentum changes by the admitted query source.

12.2 Exact finite-dimensional realization

Call a finite-dimensional stateful attention realization field-matched when its query incident is $P_R q^\sharp$, its retained state is $\mathbf{x} = (a, \dot{a})$, its generator is $\mathcal{G}_R$, and its initial state and readouts equal those of the field.

Theorem 12.1 (Exact stateful realization). *Every field-matched stateful attention realization has exactly the same instantaneous and causal responses as the finite-dimensional information field.*

Proof. Instantaneous response is $P_R q^\sharp$, with the sign convention of [theorem 7.3](#). Retained state applies the displayed impulse and propagates with $e^{\tau \mathcal{G}_R}$. Identity readout recovers a and $\dot{a}$, while D^* recovers relational field strength. Uniqueness in [theorem 9.3](#) makes both histories identical. $\square$

Equivalent construction uses the skew-adjoint generator $\mathbb{A}_D$ on $H \oplus G$. Both representations implement the same intrinsic evolution.

12.3 Response equivalence and source determination

Instantaneous attention contains only P_R , whereas causal response also contains spectral stiffness, state, and initial data. A state-space realization supplied with those same objects can represent the same causal law.

Information-field construction makes

$$R \longmapsto (P_R, D^*, L_R, E_R, K_R)$$

one coherent determination. An arbitrary state-space realization may encode equivalent operators without determining their localization, primitive relations, energy, or admissible source variations. Response equivalence therefore does not imply source equivalence.

12.4 Computational non-equivalence

Exact response equality leaves computational cost undetermined. Dense matrix exponentiation, sparse first-order exchange, spectral propagation, and electromagnetic evolution can realize the same map with different storage, arithmetic, and data movement.

Meaningful computational comparison therefore holds the input–output response, incident map, initial state, readout, and precision fixed. [Section 17](#) applies these conditions to finite response evaluation. [Section 13](#) first allows the source geometry itself to vary with the clock.

13 Time-dependent source geometry

13.1 Smooth families of relation operators

Source-changing interventions alter relational geometry rather than merely adding external forcing. Consider fixed carriers H, G and a norm- C^1 family of bounded relation operators

$$D(\tau) : G \longrightarrow H.$$

Corresponding intrinsic objects are

$$P_{R(\tau)} = P_{\text{ran } D(\tau)}, \quad L_{R(\tau)} = cD(\tau)D(\tau)^*.$$

Unbounded differential families require a common form domain and differentiability of their closed quadratic forms. Statements below use the bounded realization unless stronger hypotheses are displayed.

Temporal equation with changing geometry is

$$m\ddot{a} + L_{R(\tau)}a = P_{R(\tau)}j(\tau)^\sharp.$$

Prescribing $R(\tau)$ supplies an external structural history. Fixed-source rigidity does not determine that history.

13.2 Geometric work and loss of fixed-source closure

Instantaneous mechanical energy is

$$E_{R(\tau)}(a, \dot{a}) = \frac{m}{2}\|\dot{a}\|_H^2 + \frac{c}{2}\|D(\tau)^*a\|_G^2.$$

Proposition 13.1 (Energy balance under changing geometry). *Every regular solution satisfies*

$$\frac{d}{d\tau}E_{R(\tau)}(a, \dot{a}) = \langle \dot{a}, P_{R(\tau)}j(\tau)^\sharp \rangle_H + c\langle D(\tau)^*a, \dot{D}(\tau)^*a \rangle_G.$$

Proof. Differentiate both norm squares. Pairing the field equation with $\dot{a}$ cancels the fixed-geometry restoring term. Remaining derivative of $D(\tau)^*$ gives the geometric-work term. $\square$

Writing $u = \sqrt{c}D(\tau)^*a$ gives

$$\dot{u} = \sqrt{\frac{c}{m}}D(\tau)^*v + \sqrt{c}\dot{D}(\tau)^*a.$$

Last term is not generally determined by (v, u) . Two configurations that differ by $n \in \ker D(\tau)^*$ have the same u , yet their structural terms differ when $\dot{D}(\tau)^*n \neq 0$. A write or relation edit therefore cannot be represented as another force in the fixed-source first-order law.

13.3 Transport of active sectors

Assume constant finite rank. Projector $P_{R(\tau)}$ is then C^1 . A bounded Hilbert-space family has the same property when its ranges are closed and zero remains uniformly separated from the positive spectrum of $D(\tau)D(\tau)^*$.

Define the Kato generator

$$\mathcal{K}_R(\tau) = [\dot{P}_{R(\tau)}, P_{R(\tau)}].$$

This is Kato's parallel transport for a smooth projector family [6].

Theorem 13.2 (Kato transport of active sectors). *Generator $\mathcal{K}_R(\tau)$ is skew-adjoint and obeys*

$$\dot{P}_{R(\tau)} = [\mathcal{K}_R(\tau), P_{R(\tau)}].$$

If W_R solves

$$\dot{W}_R = \mathcal{K}_R W_R, \quad W_R(0) = I_H,$$

then $W_R(\tau)$ is orthogonal and

$$P_{R(\tau)} = W_R(\tau) P_{R(0)} W_R(\tau)^*.$$

Proof. Differentiating $P^2 = P$ gives $P\dot{P}P = 0$ and $\dot{P} = P\dot{P} + \dot{P}P$. Self-adjointness of P and $\dot{P}$ makes $[\dot{P}, P]$ skew-adjoint. Direct commutator expansion gives $[[\dot{P}, P], P] = \dot{P}$. Differentiation of $W_R^* P_{R(\tau)} W_R$ then gives zero. $\square$

Kato transport compares active fibers at different clock values. It neither supplies the source-changing intervention nor replaces the geometric-work term.

13.4 Rank transitions

Proposition 13.3 (Local constancy of projector rank). *A norm-continuous path of finite-rank orthogonal projectors has locally constant rank.*

Proof. Whenever $\|P - Q\| < 1$, restriction of Q to $\text{ran } P$ is injective. Exchanging P and Q gives the opposite rank inequality, so their ranks agree. Norm continuity supplies this bound locally. $\square$

Genuine finite-dimensional rank change therefore lies outside regular Kato transport. Smooth $D(\tau)$ may still lose rank, but its range projector cannot remain norm continuous through that event. A transition rule must state how configurations and momentum pass between the adjoining active sectors.

13.5 Junction conditions

Suppose D is piecewise regular on fixed carriers and changes at τ_0 . Let a be continuous and piecewise twice differentiable. In the absence of an impulsive source, variation of the piecewise action gives

$$a(\tau_0^-) = a(\tau_0^+), \quad m\dot{a}(\tau_0^-) = m\dot{a}(\tau_0^+).$$

Acceleration may jump because L_R , P_R , or the regular source changes. If the event data assign an active projector P_0 and an impulsive covector $q\delta_{\tau_0}$, then

$$m\dot{a}(\tau_0^+) - m\dot{a}(\tau_0^-) = P_0 q^\sharp.$$

At a discontinuous projector jump, neither one-sided projector is an event value without this assignment. Changes of carrier likewise require a supplied identification or transition map because fixed-carrier junction identities do not compare vectors in different Hilbert spaces.

13.6 Parametric excitation

Geometric work can increase or decrease field energy even when $j = 0$. Periodic variation of $D(\tau)$ modulates intrinsic stiffness and can produce parametric amplification near twice a natural frequency in a single-mode reduction.

Consider $D(\tau)D(\tau)^*$ restricted to an invariant line with stiffness

$$\lambda(\tau) = \lambda_0 [1 + \epsilon \cos(\nu\tau)].$$

Mode amplitude satisfies a Mathieu-type equation

$$m\ddot{z} + c\lambda(\tau)z = 0.$$

Growth under resonant modulation draws energy from the prescribed structural intervention and obeys [proposition 13.1](#). Cyclic return of the source to its initial geometry need not return the field to its initial state.

[Section 14](#) specializes the abstract operator to a local differential realization, where the principal symbol determines propagation speed and domains of dependence.

14 Local differential realization

14.1 Differential and regularity hypotheses

Realizing the localization algebra on a metric base and the relation operator as a local differential operator gives the field equation spatial characteristics and finite domains of dependence. These properties refine the fixed-source dynamics of [section 9](#) without altering its variational origin.

Consider a smooth Riemannian manifold M , scalar quantity carrier $H = L^2(M)$, and covector relation carrier $G = L^2(T^*M)$. On a complete boundaryless base, or after imposing a self-adjoint boundary condition, take

$$D^* = \nabla, \quad D = -\operatorname{div}, \quad DD^* = -\Delta.$$

Sign convention $\Delta = \operatorname{div} \nabla$ makes $-\Delta$ nonnegative. With admitted forcing $f = P_R j^\sharp$, the information-field equation becomes

$$m \partial_\tau^2 a - c \Delta a = f, \quad f = P_R j^\sharp.$$

Vector-bundle realizations follow under the corresponding hypotheses: a first-order operator D with smooth coefficients, a Green identity, a uniformly positive spatial principal symbol on the active sector, and boundary conditions making DD^* self-adjoint and nonnegative. Sobolev regularity supplies the traces and integrations by parts used below.

14.2 Local energy balance

For the scalar branch, define energy density and flux

$$e_R = \frac{m}{2} |\partial_\tau a|^2 + \frac{c}{2} |\nabla a|^2, \quad \mathbf{J}_R = -c(\partial_\tau a) \nabla a.$$

Proposition 14.1 (Local energy balance). *Every regular solution satisfies*

$$\partial_\tau e_R + \operatorname{div} \mathbf{J}_R = (\partial_\tau a) f.$$

Proof. Multiply the wave equation by $\partial_\tau a$ and use

$$(\partial_\tau a) \Delta a = \operatorname{div}((\partial_\tau a) \nabla a) - \langle \nabla \partial_\tau a, \nabla a \rangle.$$

Collecting clock derivatives gives the identity. □

Integration over $U \in M$ gives

$$\frac{d}{d\tau} \int_U e_R = \int_U (\partial_\tau a) f - \int_{\partial U} \mathbf{J}_R \cdot n.$$

Closed or reflecting boundary conditions annul outward flux. Open boundaries carry relational energy into or out of U . Restricting an observation to U retains the exterior source through its induced boundary flux.

14.3 Finite speed of propagation

Scalar principal part is normally hyperbolic, with characteristic speed

$$v_{\text{char}} = \sqrt{c/m}.$$

Theorem 14.2 (Finite speed of propagation). *Suppose two finite-energy scalar solutions have identical initial data and forcing inside the backward cone of a point (τ_1, p) , whose lateral boundary moves with speed v_{char} . Then their values agree at (τ_1, p) .*

Proof. Difference of the solutions obeys the same equation with zero initial energy and zero interior forcing in that cone. On its lateral boundary,

$$|\mathbf{J}_R \cdot n| \leq c |\partial_\tau a| |\nabla a| \leq v_{\text{char}} e_R.$$

Moving-boundary contribution dominates outward flux. Energy in the shrinking cone cannot become positive, so the difference vanishes at its tip. $\square$

Consequently, the local differential realization has a domain of dependence. Finite graph distance alone supplies no such cone, and a nonlocal D may couple separated regions immediately. Moreover, the theorem treats f as the admitted forcing; if $j^\sharp \mapsto P_R j^\sharp$ is nonlocal, this map introduces nonlocal dependence before the subsequent differential propagation with finite speed.

14.4 Global solutions from local data

Let $\{U_i\}$ be a locally finite cover of the relevant quantity and relation cells, or of the underlying coordinate patches. Assume complete support: every local weak equation and every boundary pairing is supported in some member of the cover. Compatible local solutions agree on overlaps in their configuration and source values, while their interface traces agree with the orientations required by the Green identity.

Theorem 14.3 (Gluing of local weak solutions). *Compatible local solutions on a cover with complete support determine a global weak solution. Conversely, every global weak solution restricts to a local solution on each member when its induced interface traces are retained. Under the energy estimate of [theorem 14.2](#), the global solution is unique for fixed initial and exterior boundary data.*

Proof. Overlap agreement defines a global configuration and source almost everywhere. Test a local weak identity against each member of a partition of unity subordinate to $\{U_i\}$, then sum the resulting identities. Oppositely oriented interior trace terms cancel by compatibility, leaving the global weak equation and the prescribed exterior boundary term. Applying the same argument after restricting a global test function proves the converse. Finally, the energy estimate applied to the difference of two solutions gives uniqueness for fixed initial and exterior boundary data. $\square$

A region lacking complete support receives interface data from every primitive relation crossing its boundary. Recomputing an isolated projector after discarding those interfaces defines a different source rather than a restriction of the same field. Finite-dimensional examples in [section 15](#) make the projector, silent sector, source variation, and distinct causal spectra explicit.

15 Finite-dimensional examples

Six finite examples isolate the distinctions established above. Projection supplies attention in the first example; relation addition changes that projection in the second; a nonzero silent direction appears in the third; equal attention projectors coexist with unequal dynamics in the fourth; doubled state recovers those dynamics in the fifth; and the final example exhibits the general source-update identities of [section 8](#).

15.1 Minimal attention response

Let $H = \mathbb{R}^2$, $G = \mathbb{R}$, and use Euclidean carrier metrics. One primitive relation places equal quantity variation at two sites,

$$D = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Active sector and projector are

$$B_R = \text{span}\{(1, 1)^\top\}, \quad P_R = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

For a query covector with Riesz representative $q^\sharp = (1, 0)^\top$, the attention response is

$$P_R q^\sharp = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Projection removes the unsupported difference direction and returns the jointly admissible component. Pairwise scoring and normalization are absent because the source geometry already determines the regulator.

15.2 Relation addition changes response

Begin with $H = \mathbb{R}^2$, $G = \mathbb{R}$, and

$$D = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad P_R = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Query representative $q^\sharp = e_2^H$ has zero response because it lies outside the active sector. Adding a second primitive relation with section e_2^H gives

$$D_{\text{add}} = I_{\mathbb{R}^2}, \quad P_{R_{\text{add}}} = I_{\mathbb{R}^2}.$$

Identical query data therefore return e_2^H . Source variation changed the geometry before evaluation; it did not add a query force.

15.3 Nonzero silent directions

For the relation in [section 15.1](#), the silent sector is

$$N_R = \ker D^* = \text{span}\{(1, -1)^\top\}.$$

Configuration $a = (1, -1)^\top$ is nonzero but has

$$F_R(a) = D^*a = 0, \quad L_Ra = 0.$$

Likewise, a nonzero query covector with representative a gives zero attention response. Silence here means orthogonality to an existing active sector, not absence of a source channel or vanishing of the incident covector.

15.4 Identical attention and distinct dynamics

Let $H = G = \mathbb{R}^2$, $c = 1$, and compare

$$D_1 = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \quad D_2 = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}.$$

Both operators are surjective, so

$$P_{R_1} = P_{R_2} = I_{\mathbb{R}^2}.$$

Every instantaneous attention response therefore agrees, whereas the intrinsic operators differ:

$$L_{R_1} = \text{diag}(1, 4), \quad L_{R_2} = \text{diag}(1, 9).$$

An impulse along e_2^H oscillates at $2/\sqrt{m}$ in the first field and $3/\sqrt{m}$ in the second. In accordance with [proposition 11.1](#), their third response jets are

$$K_{R_1}'''(0^+)e_2^H = -\frac{4}{m^2}e_2^H, \quad K_{R_2}'''(0^+)e_2^H = -\frac{9}{m^2}e_2^H.$$

Projector equality is thus insufficient for dynamical equivalence: attention sees the common active sector, while causal propagation also sees the singular values of D .

15.5 Exact state-space realization

For either source in [section 15.4](#), set $\mathbf{x} = (a, \dot{a})$ and define

$$\mathcal{G}_{R_i} = \begin{pmatrix} 0 & I_2 \\ -m^{-1}L_{R_i} & 0 \end{pmatrix}.$$

Impulse $q\delta_0$ changes doubled state by

$$\mathbf{x}(0^+) - \mathbf{x}(0^-) = \begin{pmatrix} 0 \\ m^{-1}q^\sharp \end{pmatrix}.$$

Propagation by $e^{\tau\mathcal{G}_{R_i}}$ reproduces the corresponding spectral sine and cosine solution exactly. Any realization specified only by P_{R_i} identifies the two examples. Retaining doubled state and supplying L_{R_i} instead gives the field-matched realization proved in [theorem 12.1](#).

15.6 Finite source variation

Let a finite source have active sector B_R , and let newly admitted relation columns span $\mathcal{C}_{\text{add}}$. Orthogonal residual

$$\mathcal{R}_{\text{add}} = (I_H - P_R)\mathcal{C}_{\text{add}}$$

contains precisely the new active directions. Applying [theorem 8.1](#) gives

$$P_{R_{\text{add}}} = P_R + P_{\mathcal{R}_{\text{add}}}.$$

Example in [section 15.2](#) has $\mathcal{R}_{\text{add}} = \text{span}\{e_2^H\}$. Removing that same column produces the lost subspace

$$\mathcal{L}_{\text{rem}} = \text{span}\{e_2^H\}$$

and reverses the projector update by [theorem 8.2](#). Adding a column already in B_R changes DD^* but leaves P_R fixed. Such an update preserves instantaneous attention while changing causal response, completing the finite realization of the separation theorem. [Section 16](#) constructs D and incident covectors directly from corpus observations.

16 Corpus realization from natural language

16.1 Observation law and data separation

WikiText-2 [\[7\]](#) supplies raw observations for a finite information field without questions, answers, semantic labels, pretrained representations, fitted weights, gradients, or losses. A declared observation law records directed local occurrence and its positive empirical measure.

Admitted vocabulary $\mathcal{V}$ contains terms occurring at least fifty times in the training split. Directed feature set $\mathcal{F}$ contains pairs of an offset and a neighboring term within context radius $\rho = 3$. Context windows remain within individual text records.

Only training occurrences contribute operator coefficients. Validation compares two source constructions determined from those coefficients, while test occurrences remain untouched until the construction has been selected.

16.2 Sparse corpus relation operator

Let $n(w, \varphi)$ count joint occurrences of term $w \in \mathcal{V}$ and directed feature $\varphi \in \mathcal{F}$. With

$$N_{\text{rel}} = \sum_{w, \varphi} n(w, \varphi),$$

define

$$H = \ell^2(\mathcal{V}), \quad G = \ell^2(\mathcal{F}), \quad D_{w\varphi} = \sqrt{\frac{n(w, \varphi)}{N_{\text{rel}}}}.$$

No coefficient is optimized. Positive empirical measure fixes

$$\|D\|_{\text{HS}}^2 = \sum_{w, \varphi} \frac{n(w, \varphi)}{N_{\text{rel}}} = 1.$$

Consequently, D is bounded with $\|D\| \leq 1$. Independently normalizing every term row instead gives

$$D_{w\varphi}^{\text{cond}} = \sqrt{\frac{n(w, \varphi)}{\sum_{\varphi'} n(w, \varphi')}}.$$

This conditional operator preserves each term's context shape but removes its observed marginal mass. It is therefore a different source, not a harmless numerical rescaling.

Resulting source contains 4,072 quantity coordinates, 170,676 directed relation coordinates, and 3,296,117 nonzero entries of D .

16.3 Context-induced source covectors

Concealing one occurrence leaves a nonempty directed context $C \subseteq \mathcal{F}$. Its target-independent unit relation incident is

$$q_C = \frac{1}{\sqrt{|C|}} \sum_{\varphi \in C} e_\varphi^G.$$

Concealed term does not enter this expression. Coupling this incident to relational field strength gives

$$-\langle q_C, D^*a \rangle_G.$$

Proposition 16.1 (Context-induced source covector). *Context C induces a continuous source covector $j_C \in H^*$ with*

$$j_C(a) = \langle q_C, D^*a \rangle_G = \langle Dq_C, a \rangle_H, \quad j_C^\sharp = Dq_C.$$

Moreover,

$$P_R j_C^\sharp = j_C^\sharp.$$

Proof. Boundedness gives

$$|j_C(a)| \leq \|D\| \|q_C\|_G \|a\|_H.$$

Adjointness identifies Dq_C as the Riesz representative. Membership in $\text{ran } D$ makes projection act as the identity. $\square$

Positive query-induced response at $a = 0$ is therefore

$$r_C = P_R j_C^\sharp = Dq_C.$$

Same corpus geometry thus determines both the field and the conversion of visible context into a quantity-side incident covector. No target identity, fitted query map, or subsequent projector application enters this admitted incident class.

16.4 Held-out evaluation

Coordinates of r_C are ranked only after response construction. Concealed term is used solely for scoring and never changes D , q_C , or the readout.

Validation compared the joint-measure operator with the conditional operator and a corpus-frequency control. Conditional normalization achieved 3.25 percent top-1 and 7.75 percent top-5 accuracy. Frequency alone achieved 9.95 percent and 27.35 percent. Joint-measure response achieved 15.85 percent and 34.15 percent, with mean reciprocal rank 0.2484.

Validation selected the joint-measure construction. Evaluation on 2,000 untouched test occurrences then gave

Table 7: Held-out prediction from 2,000 concealed WikiText-2 occurrences.

Readout	Top-1	Top-5	Mean reciprocal rank
Frequency control	10.15%	28.30%	—
Relational response	15.75%	34.25%	0.2539

Target response coverage was 99.25 percent. Improvement over frequency shows that directed local relation geometry contributes held-out predictive information without fitted parameters.

16.5 Field content determined by the corpus

Corpus occurrences determine the coordinates, carrier measure, relation operator, active sector, and local context incidents. At evaluation time, visible neighboring terms determine q_C , and the field relation operator determines Dq_C . Concealed terms are used solely for scoring and never enter source or incident determination.

Declared observation law resolves local lexical incidence rather than assigning semantic categories or discourse-scale reference. That scope is a property of the observations supplied to this realization, not a modification of the field law.

Corpus realization therefore establishes parameter-free source determination and nontrivial held-out response directly from natural-language occurrences. [Section 17](#) uses the resulting relation coordinates to derive exact digital response reductions.

17 Digital realization and exact response reduction

Field structure reduces computation only when the requested response preserves less information than the complete field state. Fixed geometry, declared incident and observation maps, locality, symmetry, and observation time provide such conditions. Each reduction below preserves the corresponding response exactly; none follows from a realized zero alone.

17.1 Exact response in relation coordinates

Consider a finite retrieval source whose observed prefix contains key–payload bindings. One admitted relation coordinate represents each binding,

$$De_i^G = k_i + v_{\pi(i)},$$

where k_i is the observed key coordinate and $v_{\pi(i)}$ is its associated payload coordinate. Evaluator-held payload labels do not enter D . A query for the i -th binding is the quotient coordinate

$$[e_i^G] \in G/\ker D.$$

For a fixed readout $O : H \rightarrow Z$, define the response map

$$\mathcal{M}_R = OD : G \longrightarrow Z.$$

Since $D \ker D = 0$, this map descends to $G/\ker D$ and satisfies

$$[\eta] \longmapsto \mathcal{M}_R \eta, \quad \mathcal{M}_R e_i^G = OD e_i^G.$$

Consequently, a basis query selects one stored column of $\mathcal{M}_R$. An incident supported on k relation coordinates selects and sums k columns. Ambient query construction, explicit projection, pairwise scoring, score scaling, normalization, and weighted aggregation are absent because D has already determined the admissible coupling.

For output width z and response-map rank r , indexed column application costs kz scalar multiply-adds, whereas an exact rank factorization costs $r(k+z)$. Comparing these counts before evaluation selects the cheaper exact representation.

17.2 Retrieval latency and retained state

Four associative-retrieval conditions used sequence and binding counts $(64, 4)$, $(128, 8)$, $(256, 16)$, and $(512, 64)$. Relation-coordinate response and one nonrecurrent FP16 FlashAttention call [3] both attained 1.000 retrieval accuracy in every condition. Field queries had support one, and every response map had full relation rank, so indexed column application was exact and cheaper than rank factorization.

Each latency in table 8 is the median of 200 synchronized device-resident evaluations after relation coordinates and query, key, and value arrays had been formed. Field evaluation used one fused FP16 CUDA kernel; the comparison used one FP16 scaled-dot-product-attention call forced to its FlashAttention backend. Maximum field discrepancy from full-precision relation-coordinate evaluation was 4.28×10^{-4} .

Table 8: Response latency in milliseconds. Entries of the form field/FlashAttention report both absolute times; ratios are FlashAttention time divided by field time.

Sequence / bindings	Batch times	Batch ratio	Incremental times	Incremental ratio
64 / 4	0.00616 / 0.01594	2.59×	0.00607 / 0.02207	3.64×
128 / 8	0.00722 / 0.01580	2.19×	0.00705 / 0.02210	3.13×
256 / 16	0.00730 / 0.01766	2.42×	0.00714 / 0.02397	3.36×
512 / 64	0.00736 / 0.02214	3.01×	0.00710 / 0.02295	3.23×

Relation-coordinate state retained 864, 2,752, 4,800, and 17,088 bytes across the four conditions. FlashAttention retained 8,704, 33,792, 66,560, and 131,584 bytes of prefix state, giving a 7.70× to 13.87× reduction. Field construction from the observed bindings required median times of 0.459, 0.733, 0.735, and 0.842 milliseconds; transfer required 0.115–0.133 milliseconds.

No parameter fitting or learned projection produced the relation coordinates. FlashAttention received the ambient query, key, and value representations required by its own realization. Reported ratios therefore compare correct response evaluation from relation coordinates and query, key, and value arrays; field-construction times are stated separately.

17.3 Minimal causal carrier and fixed-time response

Instantaneous response uses only P_R , whereas causal response depends on powers of L_R . For declared incident port $B_{\text{in}} : U \rightarrow H$ and readout $O : H \rightarrow Z$, set

$$\mathcal{R} = \text{span}\{L_R^k B_{\text{in}} u : k \geq 0, u \in U\}, \quad \mathcal{N}_O = \bigcap_{k \geq 0} \ker(OL_R^k).$$

Every zero-past response at these ports factors through

$$\mathcal{K}_{R,B_{\text{in}},O} = \mathcal{R} \ominus (\mathcal{R} \cap \mathcal{N}_O).$$

Self-adjointness of L_R makes this carrier and its orthogonal invisible complement invariant. Restriction preserves every response moment $OL_R^k B_{\text{in}}$, every retarded response, and every admitted forcing history. Finite reachability and observability make its dimension minimal up to similarity, as proved in [section 11](#). Nonzero initial data enter through additional position and velocity ports.

A benchmark retaining eight jointly reachable and observable coordinates inside ambient dimensions 128, 512, and 2,048 reduced modal multiplication and state storage by factors 16, 64, and 256. Median CPU response latency improved by factors 1.59, 2.12, and 3.97. Enlarging the incident port restored six observable but unreachable coordinates; enlarging the readout restored five reachable but unobservable coordinates. These interventions confirm that removal follows complete port response rather than one observed value.

When observation time t and incident history are fixed, temporal state is unnecessary. Spectral residues W_λ determine

$$T_t = \sum_{\lambda} g_t(\lambda) W_{\lambda}, \quad g_t(\lambda) = \begin{cases} \frac{1 - \cos(t\sqrt{\lambda/m})}{\lambda}, & \lambda > 0, \\ \frac{t^2}{2m}, & \lambda = 0. \end{cases}$$

Direct application of T_t replaces runtime modal propagation for that declared time. Controlled examples with 16, 64, and 256 retained modes reduced modal work by the same factors and storage by factors 9, 36, and 144. Median CPU latency improved by $12.18\times$ – $21.77\times$ for scalar response and $4.88\times$ – $26.55\times$ for batches of 256. Accelerator latency improved by $2.28\times$ – $2.51\times$ for scalar response and $2.99\times$ – $3.09\times$ for batched response.

Dense formation of D or $L_R = cDD^*$ is unnecessary. Block Krylov closure applies L_R through successive sparse actions of D^* and D . Once an orthonormal basis Q_R spans the invariant reachable carrier,

$$A_R = Q_R^* L_R Q_R$$

determines observable closure and every higher response moment. In ambient dimensions 128, 512, and 2,048, matrix-free construction used ten factorized intrinsic actions in every case and improved construction time over dense formation by factors 2.83, 16.51, and 261.31. Maximum response discrepancy was 3.47×10^{-18} .

17.4 Source variation, locality, and symmetry

Let Q_R span a previously determined reachable carrier. A changed source permits exact reconstruction within that carrier when every new incident and initial-data port lies in $\text{ran } Q_R$ and

$$L'_R \text{ran } Q_R \subseteq \text{ran } Q_R.$$

Failure of either condition requires renewed closure under L'_R . An expanding edit can begin from the union of the old basis and new port directions, close that union under the changed intrinsic operator, and then recompute reachability and observability inside the augmented carrier. This procedure preserves every new response without retaining obsolete directions.

Relation incidence supplies a separate exact reduction. Orthogonal connected components containing neither incident support nor readout support cannot enter any zero-past response. Restricting D , carrier metrics, ports, and readout to the components meeting both supports preserves complete causal response. Within a connected component, however, L_R^k is supported on walks of length at most k , while positive-time response generally contains every power of L_R . Sparse incidence therefore implies finite-order locality, not a strict finite-hop cone; strict finite propagation requires the differential hypotheses of [section 14](#).

An involutive source automorphism satisfying $U_H D = D U_G$ decomposes the causal carrier into invariant parity sectors when incident, readout, and initial-data ports respect the same involution. Separate Krylov closures then sum to the original response. This reduction becomes computationally favorable only when smaller sector widths outweigh the cost of verifying and coordinating the decomposition.

Table 9: Measured effects of exact structural reductions. Construction ratios compare reduced with complete matrix-free construction; response ratios compare equal response evaluations. Full conditions and residuals appear in [appendix J](#).

Reduction	Structural condition	Measured effect
Minimal causal carrier	Fixed incident and readout ports	$1.59\times\text{--}3.97\times$ response
Fixed-time map	Fixed time and incident history	$2.28\times\text{--}26.55\times$ response
Matrix-free closure	Sparse D , small reachable carrier	$2.83\times\text{--}261.31\times$ construction
Invariant source update	Old carrier remains invariant	$1.64\times\text{--}2.90\times$ construction
Augmented source update	Small reachable expansion	$1.25\times\text{--}2.29\times$ construction
Component restriction	Port-supporting components only	up to $3.94\times$ construction
Parity decomposition	Source and ports share an involution	$1.15\times$ at dimension 512

Symmetry was slower through dimension 256 and became faster only at dimension 512. This crossover is important: exact structural reducibility does not by itself guarantee lower wall-clock time.

17.5 Processor-independent response maps

Four finite response representations cover the fixed-source laws considered here:

1. a static relation-coordinate map for instantaneous response;
2. a fixed-time map for one declared observation time;
3. a finite family of maps for finitely many sampled times;
4. an exact modal recurrence for a regular incident history.

Source, incident and readout maps, admitted initial data, mass, forcing convention, and requested times determine each representation independently of processor architecture. Changing any determining datum requires reconstruction. Irregular histories retain interval-dependent propagation, while time-dependent source geometry retains the law of [section 13](#).

Static evaluation performs supported column reduction. Fixed-time and sampled-time evaluation apply packed linear maps. Regular-grid evaluation retains modal position and velocity, updates both under the incident history, and returns the declared observations together with final state. These operations are direct realizations of the reduced field response rather than transcriptions of neural attention operations.

17.6 Processor-specific kernels and numerical accuracy

CPU, Metal, and CUDA realizations evaluate the same four response representations. Accuracy comparisons include every admitted incident coordinate, initial position and velocity, nonzero prior states, complete recurrent histories, and returned final state. Unsupported precision is rejected rather than silently converted.

CPU FP64 evaluation differed from reference values by at most 2.22×10^{-16} . Metal and CUDA FP32 evaluation differed by at most 1.51×10^{-7} . For modal and observation widths fitting one device workgroup, one recurrent kernel updates all modes and completes readout at every step. Wider responses use one parallel modal update followed by one parallel observation kernel per step; device-stream ordering preserves the recurrence exactly.

Processor-specific recurrence kernels improved response time by $3.10\times$ – $4.40\times$ on Metal and $2.15\times$ – $7.88\times$ on CUDA over tested widths 32–128. CPU gains ranged from $1.08\times$ to $8.82\times$ before crossing in favor of the vendor tensor path. At widths 1,025, 2,048, and 4,096, the wide processor-specific kernel was faster only at width 1,025; tensor evaluation was up to $3.02\times$ faster at width 4,096. Numerical equivalence therefore permits device- and workload-specific selection without changing the field law.

17.7 Corpus source construction and response evaluation

Corpus source from [section 16](#) provides an end-to-end static case. Sparse construction retains 4,072 quantity coordinates, 170,676 relation coordinates, and 3,296,117 nonzero entries of D . A deterministic batch of 512 held-out contexts selects 1,849 relation coordinates. Identity readout applies the selected columns directly without materializing dense D , I_H , or P_R .

Loading and source determination required 5.04 seconds; sparse construction required 1.13 seconds. Reduced response took 2.84 milliseconds on CPU and 1.79 milliseconds on Metal, with 0.52 milliseconds for Metal output transfer. One complete 512-query lifecycle required 6.79 seconds on CPU and 6.74 seconds on Metal, compared with 7.91 seconds for direct evaluation of the same static response. Ten reused batches required 6.77 seconds on Metal through the reduced map and 30.63 seconds through direct evaluation. Shared source determination is charged to both paths.

Response evaluated 12,505,112 scalar multiply-adds. Held-out coverage was 99.22 percent, top-1 accuracy was 15.23 percent, and top-5 accuracy was 34.57 percent for this 512-context sample. One changed training record redetermined the source and response map in 6.90 seconds, of which 5.50 seconds belonged to source determination and 2.75 milliseconds to the replacement response.

17.8 Temporal reuse on constant-source intervals

Piecewise constant source geometry permits exact composition across intervals. Let $h_t \in \mathbb{R}^3$ be one homogeneous modal state, let Θ_e be the affine transition during interval e , and let n_e be its duration

in grid steps. Stepwise evaluation applies

$$h_{t+1} = \Theta_e h_t$$

at every step, whereas event composition applies

$$h_{t+n_e} = \Theta_e^{n_e} h_t$$

once per interval. Both paths receive identical transitions, durations, initial state, and final-state observation.

Table 10: CUDA latency in milliseconds for four constant-source intervals with 64 modes and batch size 64. Power construction is stated separately; break-even counts include that cost.

Steps / events	Stepwise	Event-composed	Ratio	Power construction	Break-even
256 / 4	0.00951	0.00637	1.49×	0.09525	31
1,024 / 4	0.01855	0.00745	2.49×	0.09040	9
4,096 / 4	0.05010	0.00726	6.90×	0.10962	3

Maximum state discrepancy was 4.96×10^{-5} , and maximum observed-response discrepancy was 2.37×10^{-5} . Event powers must be reconstructed whenever a transition or interval duration changes. Exact reuse therefore accelerates repeated response on constant-source intervals without identifying static retrieval and temporal propagation as one computational task.

Comparison conditions, measured intervals, and numerical error tolerances appear in [appendix J. Section 18](#) realizes the same finite active dynamics in a transverse electromagnetic carrier.

18 Electromagnetic realization of finite dynamics

Finite active information-field dynamics is a positive oscillator law. Every such law admits an exact realization in an invariant transverse Maxwell modal sector. Pullback along the modal encoding preserves the action, source coupling, energy, query response, and retarded response. This result realizes the information-field law electromagnetically without identifying information with electromagnetism.

18.1 Finite active-sector dynamics

Let H, G be finite dimensional and suppose $B_R = \text{ran } D \neq \{0\}$. Restriction

$$L_{R,B} = L_R|_{B_R}$$

is positive definite because $\ker L_R = B_R^\perp$. Active information-field equation is

$$m\ddot{a} + L_{R,B}a = f, \quad a, f \in B_R.$$

Modal encoding acts on this active sector. Silent directions remain in H and are annihilated by its ambient extension.

18.2 Transverse Maxwell carrier

Choose a finite disjoint union of homogeneous lossless flat periodic electromagnetic cavities. Temporal gauge and the mean-zero transverse condition define a carrier $\mathcal{T}_{\text{EM}}$. With $\mathbf{E} = -\dot{\mathbf{A}}$ and $\mathbf{B} = \text{curl } \mathbf{A}$, mass and stiffness operators are

$$M_{\text{EM}}\mathbf{A} = \epsilon\mathbf{A}, \quad K_{\text{EM}}\mathbf{A} = \text{curl}(\mu^{-1} \text{curl } \mathbf{A}).$$

Periodic integration by parts makes K_{EM} nonnegative and self-adjoint. Maxwell action on the transverse carrier is

$$\mathcal{S}_{\text{EM}}[\mathbf{A}; \mathbf{J}] = \int_{I_\tau} \left[\frac{1}{2} \langle \dot{\mathbf{A}}, M_{\text{EM}} \dot{\mathbf{A}} \rangle - \frac{1}{2} \langle \mathbf{A}, K_{\text{EM}} \mathbf{A} \rangle + \langle \mathbf{A}, \mathbf{J} \rangle \right] d\tau.$$

Selected impressed currents are transverse. Taking electric charge density zero gives

$$\text{div}(\epsilon\mathbf{E}) = 0, \quad \text{div } \mathbf{B} = 0, \quad \partial_\tau \rho_e + \text{div } \mathbf{J} = 0.$$

Periodic boundaries remove unreported surface terms.

18.3 Spectral construction of an invariant modal subspace

Theorem 18.1 (Finite transverse Maxwell realization). *A finite active information field admits a transverse Maxwell carrier, an invariant finite modal sector $V_{\text{EM}} \subset \mathcal{T}_{\text{EM}}$, and a modal encoding*

$$\mathcal{U}_{\text{EM}} : B_R \longrightarrow V_{\text{EM}},$$

and selected currents such that

$$\mathcal{U}_{\text{EM}}^* M_{\text{EM}} \mathcal{U}_{\text{EM}} = mI_{B_R}, \quad \mathcal{U}_{\text{EM}}^* K_{\text{EM}} \mathcal{U}_{\text{EM}} = L_{R,B},$$

together with the strong intertwining identity

$$K_{\text{EM}} \mathcal{U}_{\text{EM}} = M_{\text{EM}} \mathcal{U}_{\text{EM}} \frac{L_{R,B}}{m}.$$

Proof. Choose an orthonormal eigenbasis $L_{R,B}e_k = \lambda_k e_k$, with $\lambda_k > 0$, and set

$$\omega_k = \sqrt{\lambda_k/m}.$$

A reference periodic cavity mode of frequency ω_0 acquires frequency ω_0/ℓ after every length is scaled by ℓ . Choose $\ell_k = \omega_0/\omega_k$, place one selected transverse mode ϕ_k in each disjoint cavity, and normalize it by

$$\langle \phi_k, M_{\text{EM}} \phi_\ell \rangle = m\delta_{k\ell}.$$

Define

$$\mathcal{U}_{\text{EM}} a = \sum_k \langle e_k, a \rangle_H \phi_k.$$

Modal orthogonality proves both pullback identities. Individual eigenvalue equations prove the strong intertwining identity, which makes $\text{ran } \mathcal{U}_{\text{EM}}$ invariant under $M_{\text{EM}}^{-1} K_{\text{EM}}$. $\square$

18.4 Pullback of the Maxwell action and source coupling

For $f \in B_R$, define the transverse impressed current

$$J_f = \frac{1}{m} M_{\text{EM}} \mathcal{U}_{\text{EM}} f.$$

It satisfies

$$\mathcal{U}_{\text{EM}}^* J_f = f$$

and lies in the selected modal sector. Strong intertwining gives the equivalence

$$M_{\text{EM}} \mathcal{U}_{\text{EM}} \ddot{a} + K_{\text{EM}} \mathcal{U}_{\text{EM}} a = J_f \iff m \ddot{a} + L_{R,B} a = f.$$

Substitution $\mathbf{A} = \mathcal{U}_{\text{EM}} a$ into the Maxwell action gives

$$\mathcal{S}_{\text{EM}}[\mathcal{U}_{\text{EM}} a; J_f] = \int_{I_\tau} \left[\frac{m}{2} \|\dot{a}\|_H^2 - \frac{1}{2} \langle a, L_{R,B} a \rangle_H + \langle a, f \rangle_H \right] d\tau.$$

Thus the active information-field action is the pullback of the Maxwell action. Every encoded solution satisfies the full ideal Maxwell equation rather than only the restricted variational equation. Hamiltonian pullback is

$$\frac{1}{2} \langle \mathcal{U}_{\text{EM}} \dot{a}, M_{\text{EM}} \mathcal{U}_{\text{EM}} \dot{a} \rangle + \frac{1}{2} \langle \mathcal{U}_{\text{EM}} a, K_{\text{EM}} \mathcal{U}_{\text{EM}} a \rangle = \frac{m}{2} \|\dot{a}\|_H^2 + \frac{1}{2} \langle a, L_{R,B} a \rangle_H.$$

For a query q , choose

$$f_q = P_R q^\sharp, \quad J_q = J_{f_q}.$$

Pullback of the query-induced variational response is

$$-f_q = -P_R q^\sharp,$$

which is the attention response of [theorem 7.3](#).

18.5 Retarded response and readout

Define modal decoder

$$C_{\text{EM}} = \frac{1}{m} \mathcal{U}_{\text{EM}}^* M_{\text{EM}}.$$

Then

$$C_{\text{EM}} \mathcal{U}_{\text{EM}} = I_{B_R}.$$

Let $G_{\text{EM}}^+(\tau)$ denote the Maxwell retarded propagator on the selected transverse carrier. Modal closure and uniqueness give

$$C_{\text{EM}} G_{\text{EM}}^+(\tau) J_f = K_R(\tau) f.$$

Consequently, decoded electromagnetic displacement, velocity, energy, attention response, and retarded response equal their information-field counterparts. A six-dimensional information source with rank-five active sector verified pullback, intertwining, action, attention, and trajectories to residuals below 10^{-14} , with no excitation of three unselected Maxwell modes. Perturbing frequency, encoding, source, or readout broke the corresponding identity in every case.

18.6 Scope of the realization theorem

Exact realization requires finite active rank, lossless reciprocal media, scalable periodic resonators, transverse impressed currents, invariant initial data, fixed source geometry, and modal readout. Unselected Maxwell modes remain physically present but unexcited under these hypotheses.

Finite electromagnetic realization is an existence theorem for a finite active law under a declared encoding, current, clock, and readout. It neither asserts a locality-preserving embedding in one connected device nor derives a computational advantage from the electromagnetic substrate.

18.7 Measurement, calibration, and modal mismatch

Physical realization separates field development from observation. Let $x(\tau)$ denote all selected and unselected modal amplitudes and let an invertible sensor map S_{EM} produce

$$y(\tau) = S_{\text{EM}}x(\tau) + \nu(\tau).$$

Decoder calibration uses declared probe pairs $(S_{\text{EM}}x_i, C_{\text{EM}}x_i)$ spanning selected and nuisance modes. Full-rank noiseless probes determine

$$C_{\text{EM,cal}} = C_{\text{EM}}S_{\text{EM}}^{-1}.$$

Consequently, calibration can remove an invertible linear measurement distortion without modifying modal mass, stiffness, impressed current, or temporal development. Noise added to runtime measurements and noise added to calibration probes remain distinct error channels.

Physical mismatch has a different status. Replacing K_{EM} by $K_{\text{EM},\varepsilon}$ breaks strong intertwining unless the replacement preserves the required identity. Replacing J_f by $J_{f,\varepsilon}$ breaks source pullback unless $\mathcal{U}_{\text{EM}}^*J_{f,\varepsilon} = f$. Observation calibration cannot restore either missing equality.

A rank-five sensitivity experiment placed three nuisance modes beside the selected carrier. Weak sensor gain and modal mixing produced carrier RMSE 3.41×10^{-3} ; noiseless calibration reduced it to 4.01×10^{-16} . One-percent runtime measurement noise produced carrier RMSE 9.08×10^{-4} , while one-percent calibration noise produced 1.05×10^{-4} on the noiseless trajectory. One-percent modal stiffness and source-gain perturbations remained visible after calibrated measurement, with maximum trajectory errors 1.02×10^{-3} and 8.12×10^{-4} , respectively.

Normalized modal energy obeyed source-work balance to 2.45×10^{-9} over 2.4 clock units. Detailed modal construction and sensitivity calculations appear in [appendix K](#). [Section 19](#) places the realization within the hierarchy of determined structure and states the remaining empirical and mathematical boundaries.

19 Discussion

19.1 Hierarchy of determined structure

A localized relation source determines substantially more than one response value. Relation operator D determines active and silent sectors, relational field strength, intrinsic energy, static variational response, and nonnegative self-adjoint operator $L_R = cDD^*$. A supplied metric clock and the reversible variational hypotheses of [section 9](#) then determine temporal development. Local differential hypotheses further determine characteristics, energy flux, and domains of dependence.

Source determination can be nonlinear in the observations even though fixed-source evolution is linear in a . Changing admitted relations changes D , its range, singular spectrum, energy, and every

derived response. Linearity of the field equation therefore neither linearizes source construction nor identifies source variation with motion inside one configuration space.

Positive carrier measures belong to the source geometry. Corpus realization makes this point concrete: joint empirical measure gives $\|D\|_{\text{HS}} = 1$, whereas independent row normalization removes observed marginal mass and defines a different source. Improved held-out prediction from the joint-measure source shows that this distinction survives into response.

19.2 Attention and the complete response law

Attention is the query-induced difference of the static variational response at fixed source geometry,

$$\Delta_q Y_R = -P_R q^\sharp.$$

Available relations determine P_R before the query. A query supplies a covector without access to its target response, projection removes unsupported directions, and readout returns contextual information. This causal sequence performs the duty shared by neural attention mechanisms.

Attention does not exhaust the field law. Projector P_R records $\overline{\text{ran } D}$ but discards the nonzero singular values of D . Sources with identical attention response can consequently have different L_R , frequencies, response jets, and causal histories. Initial response jet

$$K'_R(0^+) = m^{-1}P_R$$

reveals active support, while later odd jets reveal powers of L_R . A realized zero may therefore express orthogonality of one incident to an existing response channel rather than absence of the channel.

Relation-coordinate incidents sharpen this interpretation. Whenever an admitted query has the form $j^\sharp = D\eta$, projector action is already implicit:

$$P_R D\eta = D\eta.$$

Query response then evaluates an intrinsic relation coordinate rather than searching an ambient collection of possible pairings.

19.3 Source variation and temporal development

Three operations remain distinct. Configuration variation changes a within one field geometry. Incident variation changes j while holding that geometry fixed. Source variation changes D , its carriers, metrics, or localization data and redetermines the geometry governing subsequent response.

Exact addition and removal identities show how finite source changes alter P_R . Smooth fixed-rank changes induce Kato transport, while rank transitions require event data at the junction. Temporal development instead propagates a configuration under the law associated with the current source. Piecewise constant sources combine both structures: each interval has fixed-source dynamics, and each event changes the law used on the next interval.

Names such as attention, memory, or gauge acquire mathematical content only after an operational criterion identifies the corresponding response or symmetry. Silence alone supplies no gauge quotient, and persistence alone supplies no storage interpretation. Such distinctions keep physical structure separate from labels imposed by an application.

19.4 Relation to neural computation

Conventional attention constructs query-conditioned coupling through learned projections, pairwise scores, masking, scaling, normalization, and weighted aggregation. Information-field response supplies the same causal duty through source-determined geometry and one variational law. Correspondence is therefore behavioral and structural rather than an operation-by-operation translation.

Several neural operations may approximate one projector or response law. Other operations may serve optimization or numerical conditioning and have no field counterpart. Conversely, silent channels, intrinsic frequencies, and causal propagation belong to the information field even when an instantaneous attention interface does not expose them.

Finite stateful attention can reproduce a finite field history when supplied with the same projector, intrinsic operator, clock, state, incident port, and readout. Exact equivalence requires all of those data. Supplying only P_R preserves instantaneous attention but cannot recover causal distinctions carried by L_R .

19.5 Computational consequences

Exact reduction follows from response structure rather than from a general claim that every field evaluation is inexpensive. Static intrinsic incidents reduce to supported columns of OD . Fixed causal ports reduce to their reachable and observable carrier. Fixed observation times reduce to Green-response maps. Disconnected components and compatible symmetry sectors reduce independently. Constant-source intervals reduce repeated temporal steps to powers of one interval transition.

Each reduction removes information that cannot affect the declared response under any admitted incident. This universal quantifier is essential. Removing a direction because one incident happens to produce zero would confuse a silent observation with an absent response channel.

Measurements in [section 17](#) show both gains and crossovers. Relation-coordinate retrieval is $2.19\times$ – $3.64\times$ faster than the matched FlashAttention response while retaining $7.70\times$ – $13.87\times$ less state. Exact causal and fixed-time reductions reach larger factors when port dimension or observation time removes most ambient modes. Symmetry and native kernels become advantageous only beyond workload-dependent crossovers. Structural reducibility thus determines what may be removed; processor behavior determines whether the resulting representation is faster at a given scale.

Corpus lifecycle measurement further separates source determination from repeated response. Natural-language occurrences determine D without fitted parameters, after which repeated intrinsic contexts reuse the same geometry. This division is the computational counterpart of the theoretical distinction between source and incident.

19.6 Substrate independence

Digital evaluation is not part of the definition of the field. Maxwell realization in [section 18](#) embeds every finite active law into an invariant transverse modal sector and preserves action, forcing, energy, attention, and retarded response. Existence of this pullback shows that the dynamical law is independent of a particular numerical architecture.

Modal encoding remains part of the realization data. Electromagnetic modes do not determine the original localization, relation typing, or readout by themselves. Likewise, invertible sensor distortion changes observed coordinates without changing dynamics, whereas stiffness or source mismatch changes the realized law before observation. Calibration can correct the former and cannot correct the latter.

19.7 Research directions

Several mathematical extensions follow directly from the present boundary:

1. identify broad observation classes that determine localized relation sources beyond finite local occurrence laws;
2. extend source determination and Kato transport to unbounded operators with changing domains and rank;
3. classify nonlinear intrinsic energies with field-dependent principal parts;
4. establish locality and finite propagation for vector bundles and noncommutative localization data;
5. characterize response laws admitting a unique typed factorization through D ;
6. quantify stability of projectors, spectra, and incident covectors under noisy source observations.

Empirical extensions should test source determination from nonlocal relations, reference structure, compositional incidents, and changing corpora while preserving target independence. Generative evaluation requires a declared readout and realistic vocabulary, context, and decoding. Physical extensions require connected realizations, measured loss and disorder, source and sensor engineering, and laboratory energy measurements. In every case, separating source determination, incident determination, field response, readout, and realization cost identifies which scientific claim an observed failure bears on.

20 Conclusion

Localized quantities, positive carrier metrics, primitive relations, and a closed relation operator determine an information field. Source $R = (A, H, G, D)$ fixes active sector $B_R = \overline{\text{ran } D}$, silent sector $N_R = \ker D^*$, field strength D^*a , intrinsic energy, and static variational response

$$Y_R(a; j) = cDD^*a - P_R j^\sharp.$$

A raw query determines one faithful incident covector precisely when all admitted relational readings lie in one affine class modulo $\ker D$. Every query specified independently of its target response then changes the variational response by

$$-P_R q^\sharp,$$

which performs attention after readout. Intrinsic incidents $q^\sharp = D\eta$ already lie in the active sector, so their response requires no separate projector evaluation.

A supplied metric clock and the closed reversible variational hypotheses determine, up to positive calibrations and temporal boundary terms,

$$m\ddot{a} + cDD^*a = P_R j^\sharp.$$

This law distinguishes instantaneous attention from complete causal response. Projector P_R records active support; operator $L_R = cDD^*$ additionally records intrinsic frequencies, later response jets, and propagation. Source variation redetermines both objects, while temporal development propagates configurations under the law associated with the current source.

Finite realizations establish the resulting consequences. Raw WikiText-2 occurrences determine a sparse source and held-out response without learned parameters. Relation-coordinate retrieval attains the same accuracy as the matched FlashAttention comparison with $2.19\times$ – $3.64\times$ lower response time and $7.70\times$ – $13.87\times$ less retained state. Fixed ports, observation times, connected components, compatible symmetries, and constant-source intervals yield further exact reductions under their stated hypotheses.

Every finite active information-field law also admits an invariant transverse Maxwell realization preserving action, forcing, energy, attention, and retarded response. Information therefore acquires a source-determined geometry, variational response, and causal dynamics; attention is its query-restricted static response.

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A Closed operators and orthogonal decompositions

Representation theory for closed quadratic forms identifies the strong operator domain, orthogonal decomposition identifies the active and silent sectors, and closed-range theory supplies the coercive estimates and weak–strong response distinction used by the field geometry and energy.

A.1 Closed forms and operator domains

Let $D : \text{Dom}(D) \subseteq G \rightarrow H$ be densely defined and closed. Its adjoint D^* is densely defined and closed. Quadratic form

$$\mathfrak{q}_R(a, \xi) = c \langle D^* a, D^* \xi \rangle_G, \quad a, \xi \in \text{Dom}(D^*)$$

is closed and nonnegative under the graph norm

$$\|a\|_{\mathfrak{q}_R}^2 = \|a\|_H^2 + c \|D^* a\|_G^2.$$

Proposition A.1 (Operator represented by the closed form). *Operator $L_R = cDD^*$ is the self-adjoint operator represented by $\mathfrak{q}_R$, with*

$$\text{Dom}(L_R) = \{a \in \text{Dom}(D^*) \mid D^* a \in \text{Dom}(D)\}.$$

Proof. Closed-form representation gives one nonnegative self-adjoint operator T . Membership $a \in \text{Dom}(T)$ means that some $b \in H$ satisfies

$$c \langle D^* a, D^* \xi \rangle_G = \langle b, \xi \rangle_H$$

for every $\xi \in \text{Dom}(D^*)$. Definition of the adjoint of D^* makes this equivalent to $D^* a \in \text{Dom}(D^{**}) = \text{Dom}(D)$, with $b = cDD^* a$. Hence $T = L_R$. $\square$

Form domain equals $\text{Dom}(L_R^{1/2}) = \text{Dom}(D^*)$, with

$$\|L_R^{1/2} a\|_H^2 = c \|D^* a\|_G^2.$$

A.2 Active and silent decomposition

Proposition A.2 (Active and silent decomposition). *Quantity carrier H decomposes orthogonally as*

$$H = \overline{\text{ran } D} \oplus \ker D^*.$$

Proof. A vector $a \in H$ is orthogonal to $\text{ran } D$ exactly when

$$\langle D\eta, a \rangle_H = 0$$

for every $\eta \in \text{Dom}(D)$. This is precisely the adjoint condition $a \in \text{Dom}(D^*)$ with $D^* a = 0$. Taking orthogonal complements and closing the range proves the decomposition. $\square$

Nonnegativity gives

$$\ker L_R = \ker D^* = N_R.$$

Spectral theorem therefore yields

$$P_R = \mathbf{1}_{(0,\infty)}(L_R), \quad P_{N_R} = \mathbf{1}_{\{0\}}(L_R).$$

No closed-range hypothesis is needed for either orthogonal projector.

A.3 Closed-range conditions

Proposition A.3 (Closed-range criterion). *Closedness of $\text{ran } D$ is equivalent to closedness of $\text{ran } D^*$. It is also equivalent to the existence of $\gamma > 0$ such that*

$$\|D\eta\|_H \geq \gamma\|\eta\|_G \quad \text{for } \eta \in \text{Dom}(D) \cap (\ker D)^\perp.$$

Equivalent quantity-side estimate is

$$\|D^*a\|_G \geq \gamma\|a\|_H \quad \text{for } a \in \text{Dom}(D^*) \cap N_R^\perp.$$

Proof. Assume first that $\text{ran } D$ is closed. Restriction of D to $\text{Dom}(D) \cap (\ker D)^\perp$ is a closed bijection onto $\text{ran } D$. Its inverse is bounded by the closed graph theorem, giving the first estimate. Conversely, if the first estimate holds and $D\eta_n$ converges in H , remove the components of η_n in $\ker D$. Resulting sequence is Cauchy in G ; closedness of D places its limit in $\text{Dom}(D)$ and the limit of $D\eta_n$ in $\text{ran } D$. Thus the range is closed. Hilbert-space closed-range duality gives closedness of $\text{ran } D^*$ and the equivalent adjoint estimate. $\square$

Such a gap isolates zero from the positive spectrum on the active sector and supplies bounded inverse estimates. Uniform transport and bounded-minimizer arguments may impose this estimate; attention $P_R q^\sharp$ itself requires no closed-range hypothesis.

A.4 Weak and strong variation

For $a, \xi \in \text{Dom}(D^*)$ and continuous $j \in H^*$, weak variation is

$$\delta V_R(a; j)[\xi] = \mathfrak{q}_R(a, \xi) - j(P_R \xi).$$

Boundedness of P_R on the form domain makes this variation continuous even when $a \notin \text{Dom}(L_R)$. An H -valued representative exists precisely when $a \in \text{Dom}(L_R)$, in which case

$$Y_R(a; j) = L_R a - P_R j^\sharp.$$

Conflating these domains would incorrectly apply DD^* to every finite-energy configuration.

B Weighted-incidence operators and finite source variations

Finite coordinates inherit the carrier metrics rather than the Euclidean inner product. Adjoint, orthogonality, pseudoinverse, and projector must therefore be taken with respect to those metrics.

B.1 Metric adjoint

Finite carriers use coordinate metrics

$$\langle a, b \rangle_H = a^\top M_H b, \quad \langle \eta, \zeta \rangle_G = \eta^\top W_G \zeta,$$

where M_H, W_G are positive diagonal matrices. Coordinate relation matrix has metric adjoint

$$D^* = W_G^{-1} D^\top M_H.$$

Indeed,

$$\langle D\eta, a \rangle_H = \eta^\top D^\top M_H a = \langle \eta, W_G^{-1} D^\top M_H a \rangle_G.$$

Euclidean transpose equals the Hilbert adjoint only when both metric matrices are identity.

B.2 Weighted orthogonal projector

Introduce the Euclidean whitening

$$\tilde{D} = M_H^{1/2} D W_G^{-1/2}.$$

Metric Moore–Penrose inverse is

$$D^+ = W_G^{-1/2} \tilde{D}^+ M_H^{1/2},$$

where $\tilde{D}^+$ is the ordinary Euclidean Moore–Penrose inverse. Orthogonal projectors onto $\text{ran } D \subseteq H$ and $(\ker D)^\perp \subseteq G$ are

$$P_R = D D^+, \quad Q_R = D^+ D.$$

When D has full column rank,

$$P_R = D(D^* D)^{-1} D^*.$$

Rank-deficient sources require D^+ or an orthonormal range basis. Direct inversion of $D^* D$ is then undefined; regularization defines a different operator and generally a different projector.

B.3 Projector update under relation addition

Suppose new columns span $\mathcal{C}_{\text{add}}$. Their old-active and orthogonal residual parts satisfy

$$\mathcal{C}_{\text{add}} \subseteq B_R + \mathcal{R}_{\text{add}}, \quad \mathcal{R}_{\text{add}} = (I_H - P_R) \mathcal{C}_{\text{add}}.$$

Subspaces B_R and $\mathcal{R}_{\text{add}}$ are orthogonal, and

$$B_{R_{\text{add}}} = B_R \oplus \mathcal{R}_{\text{add}}.$$

Hence

$$P_{R_{\text{add}}} = P_R + P_{\mathcal{R}_{\text{add}}}.$$

Projector does not change when every new column lies in B_R . Intrinsic operator generally does change because each added relation coordinate contributes a positive rank-one term whose coefficient is fixed by its relation-carrier metric.

B.4 Projector update under relation removal

After removing declared columns, updated range satisfies $B_{R_{\text{rem}}} \subseteq B_R$. Define

$$\mathcal{L}_{\text{rem}} = B_R \cap B_{R_{\text{rem}}}^\perp.$$

Orthogonal decomposition

$$B_R = B_{R_{\text{rem}}} \oplus \mathcal{L}_{\text{rem}}$$

gives

$$P_{R_{\text{rem}}} = P_R - P_{\mathcal{L}_{\text{rem}}}.$$

Removing a redundant column may leave the range unchanged while changing stiffness. Removing a range-essential column changes both support and stiffness.

B.5 Rank transitions

Finite addition changes rank by

$$\text{rank } D_{\text{add}} - \text{rank } D = \dim \mathcal{R}_{\text{add}}.$$

Finite removal changes rank by

$$\text{rank } D - \text{rank } D_{\text{rem}} = \dim \mathcal{L}_{\text{rem}}.$$

These identities separate source edits that merely reweight existing active directions from edits that create or delete response channels. [Appendix C](#) proves the rigidity of the minimal quadratic energy built from the resulting field strength.

C Rigidity of the intrinsic energy

Minimal first-order energy depends on a configuration only through its relational field strength and the metric of the relation carrier. Full orthogonal invariance then leaves one positive calibration and no additional constitutive operator.

C.1 Admissible quadratic class

After $g = D^*a$ has been formed, consider a continuous quadratic functional $\Phi_G : G \rightarrow \mathbb{R}$ that uses only the carrier metric and satisfies

$$\Phi_G(0) = 0, \quad \Phi_G(Ug) = \Phi_G(g)$$

for every orthogonal U on G . No further occurrence of D , D^* , or a localization operator is admitted in this minimal first-order class. Assume nonnegativity and strict positivity for nonzero g . Riesz representation gives one bounded self-adjoint T_G such that

$$\Phi_G(g) = \frac{1}{2} \langle g, T_G g \rangle_G.$$

C.2 Rigidity under orthogonal invariance

Lemma C.1 (Orthogonal-invariance rigidity). *If a bounded self-adjoint operator commutes with every orthogonal operator on a real Hilbert space, then it is a scalar multiple of the identity.*

Proof. Commutation with reflection across the orthogonal complement of a unit vector e makes $\text{span}\{e\}$ invariant. Thus every unit vector is an eigenvector. For orthogonal unit vectors e, f , linearity applied to $e + f$ forces their eigenvalues to agree. Repeating over two-dimensional spans gives one scalar on the complete carrier. $\square$

Invariance of Φ_G gives $U^*T_GU = T_G$, so the lemma yields $T_G = cI_G$. Strict positivity gives $c > 0$.

C.3 Pullback to configurations

Substitution $g = D^*a$ gives the unique energy in this class,

$$V_R(a; 0) = \frac{c}{2} \|D^*a\|_G^2.$$

Source naturality follows from $D'^*U_H = U_GD^*$. Orthogonally isomorphic sources preserve the norm, so

$$V_{R'}(U_H a; 0) = V_R(a; 0).$$

C.4 Residual calibration freedom

Positive scalar c is a calibration because source data contain no preferred energy unit. Reapplying D or D^* to the field strength defines higher-relational-order energies outside the declared minimal class. A supplied positive operator Γ permits

$$\frac{1}{2} \langle D^*a, \Gamma D^*a \rangle_G.$$

Declared relation types may reduce the orthogonal symmetry and permit different coefficients on invariant sectors. Nonquadratic dependence may produce field-dependent principal parts. Each extension requires additional constitutive data or a broader admissible class. [Appendix D](#) applies the same rigidity principle to temporal dynamics.

D Rigidity and well-posedness of temporal dynamics

Static energy admits inequivalent temporal laws. Reversibility, source naturality, metric isotropy, first-jet locality, and a field-independent second-order principal part select the inertial law within the class stated in [section 9](#).

D.1 Non-uniqueness from static energy

On a positive eigenmode $L_R e = \lambda e$, source-free gradient development and inertial development reduce to

$$\dot{z} + \lambda z = 0, \quad m\ddot{z} + \lambda z = 0.$$

Their respective solutions decay and oscillate. Both share the stationary condition $\lambda z = 0$. Static source data therefore cannot distinguish them.

Gradient branch additionally assumes a force-to-rate map and breaks time-reversal invariance. Such structure is absent from the closed reversible variational class.

D.2 First-jet classification

At one clock point, let $v = \dot{a}$ and $g = D^*a$. Orthogonal invariance on the configuration carrier makes the field-independent positive kinetic term

$$\frac{m}{2} \|v\|_H^2.$$

with $m > 0$. A nonscalar positive inertia operator is independent constitutive data. Orthogonal invariance makes the field-strength density a function $\psi(r)$ of $r = \|g\|_G^2/2$. Its Hessian is

$$\psi'(r)I_G + \psi''(r)g \otimes g.$$

A field-independent second-order principal part forces $\psi'' = 0$, hence $\psi(r) = cr + b$. Stability gives $c > 0$.

No primitive pairing between $v \in H$ and $g \in G$ exists. Source-derived mixed term is

$$\langle D^*\dot{a}, D^*a \rangle_G = \frac{1}{2} \frac{d}{d\tau} \|D^*a\|_G^2,$$

which is a boundary derivative. Additional dependence on $D^*\dot{a}$ replaces the field-independent kinetic principal part by one involving DD^* . Time-reversal invariance excludes remaining odd-velocity terms.

D.3 Sign and Euler–Lagrange equation

Kinetic-minus-potential sign gives positive Legendre energy. Opposite potential sign yields

$$m\ddot{a} - L_R a = 0$$

and produces $\cosh(\tau\sqrt{\lambda/m})$ on positive spectral modes. Unbounded L_R then prevents bounded evolution on the natural energy carrier.

Compactly supported variation of the closed-form action

$$\mathcal{S}_R[a; j] = \int \left[\frac{m}{2} \|\dot{a}\|_H^2 - \frac{c}{2} \|D^*a\|_G^2 + j(P_R a) \right] d\tau$$

gives the weak equation, and its strong representative on $\text{Dom}(L_R)$ is

$$m\ddot{a} + L_R a = P_R j^\sharp.$$

D.4 Spectral Cauchy solution

Let E_{L_R} be the projection-valued spectral measure of L_R . For initial data $(a_0, v_0) \in \mathcal{H}_R^{\text{en}}$, define

$$a(\tau) = \int_{[0, \infty)} \cos(\tau\sqrt{\lambda/m}) dE_{L_R}(\lambda) a_0 + \int_{[0, \infty)} \frac{\sin(\tau\sqrt{\lambda/m})}{\sqrt{\lambda/m}} dE_{L_R}(\lambda) v_0,$$

with the sine quotient equal to τ at $\lambda = 0$. Boundedness of cosine and sinc multipliers on finite clock intervals proves finite-energy existence and continuous dependence. Scalar uniqueness on almost every spectral fiber proves uniqueness. Duhamel integration treats forcing in $L_{\text{loc}}^1(I_\tau; H)$.

For source-free evolution, differentiation under bounded spectral truncation conserves

$$\frac{m}{2}\|\dot{a}\|_H^2 + \frac{1}{2}\|L_R^{1/2}a\|_H^2.$$

Limit in the graph-energy topology extends conservation to every finite-energy weak solution. Conserved mechanical energy is a seminorm on the initial-position component whenever $N_R \neq \{0\}$, while the graph norm also controls that silent position. [Appendix E](#) realizes the same dynamics as skew-adjoint first-order exchange.

E First-order generators and polar decomposition

Quantity configurations and primitive relations form two sectors of a single conservative exchange. A skew-adjoint generator on their direct sum gives the first-order evolution, while its compatible configuration-generated sector recovers the second-order information-field equation. Polar decomposition then separates the common positive spectrum from the distinct null sectors.

E.1 Skew-adjoint block operator

For $v \in H$ and $u \in G$, define

$$\mathbb{A}_D(v, u) = (-Du, D^*v)$$

on $H \oplus G$, with domain $\text{Dom}(D^*) \oplus \text{Dom}(D)$. For two domain vectors,

$$\langle \mathbb{A}_D(v, u), (x, y) \rangle_{H \oplus G} = -\langle u, D^*x \rangle_G + \langle v, Dy \rangle_H = -\langle (v, u), \mathbb{A}_D(x, y) \rangle_{H \oplus G}.$$

Thus $\mathbb{A}_D$ is skew-symmetric. Suppose (x, y) belongs to its adjoint domain. Boundedness of the adjoint functional first with $v = 0$ and then with $u = 0$ forces $x \in \text{Dom}(D^*)$ and $y \in \text{Dom}(D^{**}) = \text{Dom}(D)$, respectively. Conversely, every such (x, y) defines a bounded adjoint functional, and its representing vector is $(Dy, -D^*x)$. Hence

$$\mathbb{A}_D^* = -\mathbb{A}_D.$$

Skew-adjointness generates an orthogonal one-parameter group and proves source-free first-order well-posedness directly.

E.2 Compatibility and second-order reduction

Scaled variables

$$v = \sqrt{m}\dot{a}, \quad u = \sqrt{c}D^*a$$

obey

$$\partial_\tau(v, u) = \sqrt{c/m}\mathbb{A}_D(v, u) + (P_R j^\sharp / \sqrt{m}, 0).$$

Whenever $\dot{a} = v/\sqrt{m}$, the second first-order equation gives

$$\partial_\tau(u - \sqrt{c}D^*a) = \sqrt{c/m}D^*v - \sqrt{c}D^*\dot{a} = 0.$$

Initial compatibility $u = \sqrt{c}D^*a$ therefore defines an invariant subspace, and the first equation on that subspace is $m\ddot{a} + cDD^*a = P_R j^\sharp$. Squaring the generator gives

$$\mathbb{A}_D^2 = -\text{diag}(DD^*, D^*D).$$

Its operator domain is

$$\text{Dom}(\mathbb{A}_D^2) = \text{Dom}(DD^*) \oplus \text{Dom}(D^*D).$$

This identity places quantity-side and relation-side second-order laws inside one first-order exchange.

E.3 Polar decomposition

Closed densely defined D has polar decomposition

$$D = U_D |D|, \quad |D| = (D^*D)^{1/2}.$$

Initial and final spaces of U_D are

$$\overline{\text{ran } D^*} \quad \text{and} \quad \overline{\text{ran } D},$$

respectively. Therefore

$$U_D^* U_D = Q_R, \quad U_D U_D^* = P_R.$$

Here Q_R is the relation-side orthogonal projector onto $\overline{\text{ran } D^*} = (\ker D)^\perp$. As an identity of nonnegative self-adjoint operators on their natural domains,

$$DD^* = U_D |D|^2 U_D^*.$$

Restriction of U_D to $(\ker D)^\perp$ is unitary onto $\overline{\text{ran } D}$. It therefore intertwines the positive spectral restrictions of D^*D and DD^* , including their spectral multiplicities, but does not identify their null spaces.

E.4 Null sectors and Fredholm index

Kernel of the block generator is

$$\ker \mathbb{A}_D = \ker D^* \oplus \ker D.$$

First summand contains silent quantity configurations. Second contains relation-coordinate redundancies. Compatible configuration-generated states contain no component in $\ker D$ because $\text{ran } D^* \perp \ker D$.

For a closed Fredholm operator D , its index

$$\text{ind } D = \dim \ker D - \dim \ker D^*$$

is locally constant under gap-continuous perturbations through closed Fredholm operators. Equivalently, the bounded transform $D(I_G + D^*D)^{-1/2}$ has locally constant index in the norm topology on bounded Fredholm operators. Such an index counts the graded null imbalance but supplies no gauge quotient by itself. Appendix F determines which parts of this generator can be recovered from causal input–output response.

F Causal response and identifiability

Causal response reveals more of a source than instantaneous attention without recovering every source datum. Calibrated response jets determine the active projector and intrinsic operator; fixed ports expose only their reachable and observable restriction; even a minimal restriction is unique only up to similarity. Retarded-kernel construction, response-jet identities, and minimal-realization theory establish these identifiability statements.

F.1 Retarded kernel

For an impulse $f\delta_0$ with $f \in B_R$, the scalar equation on a positive spectral subspace of L_R is

$$m\ddot{z} + \lambda z = f\delta_0.$$

Continuity of z and integration across zero give $m[\dot{z}]_0^{0+} = f$. Its zero-past solution is

$$z(\tau) = \mathbf{1}_{[0,\infty)}(\tau) \frac{\sin(\tau\sqrt{\lambda/m})}{m\sqrt{\lambda/m}} f.$$

Since $B_R = (\ker L_R)^\perp$, no zero-frequency component is excited by f . Spectral integration gives

$$K_R(\tau) = \mathbf{1}_{[0,\infty)}(\tau) \frac{1}{m} \Omega_R^{-1} \sin(\tau\Omega_R) P_R.$$

F.2 Response jets

For bounded L_R , the sine series converges in operator norm and gives

$$K_R^{(2k+1)}(0^+) = \frac{(-1)^k}{m^{k+1}} L_R^k P_R.$$

For unbounded L_R , spectral truncation gives the same strong right derivative on $\text{Dom}(L_R^k)$. Thus the first jet recovers P_R/m , while the third recovers $-L_R/m^2$. No response jet selects a unique factor D from DD^* : relation-side unitary freedom and invisible relation-null directions remain.

F.3 Reciprocity

For every $\tau > 0$, $K_R(\tau)$ is a real Borel function of L_R multiplied by the commuting spectral projector P_R . Hence

$$K_R(\tau)^* = K_R(\tau).$$

For any source and observation vectors x, y ,

$$\langle x, K_R(\tau)y \rangle_H = \langle y, K_R(\tau)x \rangle_H.$$

Intrinsic closed response is therefore reciprocal. Antisymmetric transport, circulation, or one-way amplification does not follow from this self-adjoint source law.

F.4 Reachable and observable response

Assume H is finite dimensional. An admitted incident class $\mathcal{J}_0 \subseteq H^*$ reaches

$$\mathcal{R}_{R,\mathcal{J}_0} = \text{span}\{L_R^k P_R j^\sharp \mid k \geq 0, j \in \mathcal{J}_0\}.$$

Readout O cannot observe

$$\mathcal{N}_{R,O} = \bigcap_{k \geq 0} \ker(OL_R^k).$$

Self-adjointness makes both subspaces invariant under L_R . Transfer data therefore determine at most

$$\mathcal{R}_{R,\mathcal{J}_0} / (\mathcal{R}_{R,\mathcal{J}_0} \cap \mathcal{N}_{R,O}).$$

Cayley–Hamilton reduces both intersections to powers $0 \leq k < \dim H$.

F.5 Minimal realization and identifiability

Suppose two finite first-order realizations (A_i, B_i, C_i) , $i = 1, 2$, have identical transfer functions and are both controllable and observable. Equality of transfer functions implies $C_1 A_1^k B_1 = C_2 A_2^k B_2$ for every $k \geq 0$. Define

$$T \left(\sum_k A_1^k B_1 z_k \right) = \sum_k A_2^k B_2 z_k.$$

If the vector on the left vanishes, applying $C_1 A_1^\ell$ and using equality of the Markov parameters shows that the vector on the right is invisible to $C_2 A_2^\ell$ for every ℓ . Observability of the second realization makes that vector zero, so T is well defined. Reversing the argument gives injectivity, while controllability gives surjectivity. Consequently, $TA_1 = A_2 T$, $TB_1 = B_2$, and $C_1 = C_2 T$.

Similarity need not be orthogonal, preserve carrier metrics, respect the $H \oplus G$ grading, or factor through isometries satisfying $U_H D = D' U_G$. Transfer equality therefore establishes uniqueness of a minimal input–output realization up to similarity, not recovery of the localized relation source.

F.6 Spectral residues and fixed-time response

Finite spectral calculus gives

$$Of(L_R)B_{\text{in}} = \sum_{\lambda} f(\lambda) O E_{\lambda} B_{\text{in}}.$$

Consequently, every fixed-port response factors through residues $W_{\lambda} = O E_{\lambda} B_{\text{in}}$. Restriction to one eigenspace turns all powers of L_R into scalar multiples. Reachable directions modulo the readout kernel therefore have dimension $\text{rank } W_{\lambda}$. Summation over orthogonal eigenspaces proves [proposition 11.5](#).

For constant forcing and zero initial data, spectral integration gives the fixed-time response

$$T_t = \sum_{\lambda > 0} \frac{1 - \cos(t\sqrt{\lambda/m})}{\lambda} W_{\lambda} + \frac{t^2}{2m} W_0.$$

Active ports satisfy $\text{ran } B_{\text{in}} \subseteq B_R$ and hence $W_0 = 0$. Composing the exact one-interval propagator treats piecewise-constant forcing on a regular grid. Distinct continuous-time pole pairs contribute twice their residue rank to McMillan degree, but sampling may alias them and lower the discrete degree.

F.7 Symmetry reduction under invariant input and output maps

Orthogonal source automorphism (U_H, U_G) obeying $U_H D = D U_G$ gives $U_H L_R U_H^* = L_R$. For involutive U_H , projectors $\Pi_{\pm} = (I_H \pm U_H)/2$ therefore commute with every Borel function of L_R .

Let E collect incident and declared initial-state ports. Invariance of $\text{ran } E$ gives

$$\Pi_{\pm} \mathcal{R}_{R,E} \subseteq \mathcal{R}_{R,E}.$$

Invariance of $\text{ran } O^*$ gives invariance of $\ker O$. Consequently, each $\ker(OL_R^k)$, their intersection, and the reachable-and-observable orthogonal complement are parity invariant. This proves [proposition 11.6](#).

Dropping port invariance invalidates the conclusion. For example, take $L_R = \lambda I$, let U_H exchange two coordinates, and let both incident and readout select only the first coordinate. Although L_R is symmetric under the exchange, its one-dimensional minimal carrier is not invariant. Splitting the selected coordinate into exchange parities produces two directions rather than one.

F.8 Conservation laws and gauge directions

On the real quantity carrier, let $J^* = -J$, $[J, L_R] = 0$, and $m\ddot{a} + L_R a = 0$. Differentiation yields

$$\frac{d}{d\tau} m \langle \dot{a}, Ja \rangle_H = -\langle L_R a, Ja \rangle_H + m \langle \dot{a}, J\dot{a} \rangle_H = 0.$$

Forcing replaces zero on the right by $\langle f, Ja \rangle_H$. Conservation therefore depends on source incidence and does not license a quotient.

A declared gauge tangent g satisfying $D^*g = 0$, $Og = 0$, and $g \perp \text{ran } B_{\text{in}}$ is absent from zero-state transfer behavior. Satisfaction of $D^*g = 0$ alone still permits observation through initial position or velocity. Likewise, $h \in \ker D$ is a redundant relation direction rather than a quantity-side gauge tangent.

F.9 Arithmetic lower bounds for response evaluation

Suppose a finite response map $\mathcal{L}$ factors as AB through a finite-dimensional carrier W . Rank inequalities give

$$\text{rank } \mathcal{L} \leq \min\{\text{rank } A, \text{rank } B\} \leq \dim W.$$

Truncated singular-value factorization at rank $\mathcal{L}$ attains equality.

For coordinate-access execution, let $\mathcal{L}e_j \neq 0$. Inputs u and $u + \alpha e_j$ agree away from coordinate j and produce distinct outputs for some $\alpha \neq 0$. Any deterministic exact algorithm must therefore inspect coordinate j on at least one worst-case input. A dual statement applies to nonzero output rows when the output is explicitly materialized.

For discrete LTI data (A, B, C) , factorization of the block Hankel operator through reachable and observable state gives rank at most state dimension. Minimal reachable-and-observable realization attains its stabilized rank. Cayley–Hamilton bounds stabilization by n block rows and columns for a supplied n -state realization.

G Time-dependent source geometry

Changing relational structure performs work on the field and moves its active sector. Smooth constant-rank changes admit canonical Kato transport, whereas rank changes and discontinuous source events require junction data. Energy differentiation, projector transport, and rank discontinuity yield the geometric-work identity, transport law, obstruction, and junction conditions.

G.1 Geometric work

Let $D(\tau) : G \rightarrow H$ be norm C^1 , let j be continuous, and let $a \in C^2(I_\tau; H)$ solve

$$m\ddot{a} + cD(\tau)D(\tau)^*a = P_{R(\tau)}j(\tau)^\sharp.$$

Energy differentiation gives

$$\begin{aligned} \frac{d}{d\tau} \left[\frac{m}{2} \|\dot{a}\|_H^2 + \frac{c}{2} \|D(\tau)^*a\|_G^2 \right] &= \langle \dot{a}, P_{R(\tau)}j(\tau)^\sharp \rangle_H \\ &\quad + c \langle D(\tau)^*a, \dot{D}(\tau)^*a \rangle_G. \end{aligned}$$

Indeed, pairing the field equation with $\dot{a}$ cancels $c\langle \dot{a}, D(\tau)D(\tau)^*a \rangle_H$; differentiating $D(\tau)^*$ leaves the second term, which measures work performed by structural change. With $v = \sqrt{m}\dot{a}$, the product rule also gives

$$\partial_\tau(\sqrt{c}D(\tau)^*a) = \sqrt{c/m}D(\tau)^*v + \sqrt{c}\dot{D}(\tau)^*a.$$

Dependence on the full configuration prevents closure in $(v, u) = (\sqrt{m}\dot{a}, \sqrt{c}D(\tau)^*a)$ unless another condition determines the missing component in $\ker D(\tau)^*$.

G.2 Kato transport

For a finite-dimensional norm- C^1 , constant-rank family, the range projector is C^1 . Set

$$\mathcal{K}_R = [\dot{P}_R, P_R].$$

Differentiating $P_R^2 = P_R$ gives

$$P_R\dot{P}_R P_R = 0, \quad \dot{P}_R = P_R\dot{P}_R + \dot{P}_R P_R.$$

Self-adjointness of P_R and $\dot{P}_R$ then gives

$$\mathcal{K}_R^* = -\mathcal{K}_R, \quad [\mathcal{K}_R, P_R] = \dot{P}_R.$$

Let $W_R(0) = I_H$ and $\dot{W}_R = \mathcal{K}_R W_R$. Skew-adjointness of $\mathcal{K}_R$ preserves $W_R^* W_R$, so W_R is orthogonal. Moreover,

$$\frac{d}{d\tau} (W_R^* P_R W_R) = W_R^* (\dot{P}_R - [\mathcal{K}_R, P_R]) W_R = 0.$$

Therefore

$$P_{R(\tau)} = W_R(\tau) P_{R(0)} W_R(\tau)^*.$$

Norm- C^1 bounded Hilbert-space families admit the same conclusion when zero is uniformly separated from the positive spectrum of $D(\tau)D(\tau)^*$. Riesz spectral calculus then makes $P_{R(\tau)}$ norm C^1 , after which the same differential argument applies.

G.3 Rank-change obstruction

If orthogonal projectors P, Q satisfy $\|P - Q\| < 1$, then $Q|_{\text{ran } P}$ is injective. Indeed, $Qx = 0$ and $Px = x$ would imply

$$\|x\| = \|(P - Q)x\| \leq \|P - Q\|\|x\|,$$

and hence $x = 0$. Reversing the roles of P and Q gives equal finite ranks. Norm-continuous finite-rank projector paths therefore have locally constant rank.

Every genuine finite-rank transition consequently breaks norm continuity of the projector. Orthogonal Kato transport cannot cross the transition because orthogonal conjugation preserves rank.

G.4 Junction conditions

For fixed carriers and a piecewise regular source, take continuous admissible paths that are regular on either side of τ_0 . Integration by parts in the temporal action yields the internal boundary contribution

$$\langle m\dot{a}(\tau_0^-) - m\dot{a}(\tau_0^+), \xi(\tau_0) \rangle_H.$$

Configuration continuity belongs to the admitted path class. Arbitrariness of the common boundary variation gives momentum continuity when no impulse is present. An impulse $q\delta_{\tau_0}$ contributes $\langle P_0 q^\sharp, \xi(\tau_0) \rangle_H$, where P_0 is the projector assigned to the event, and changes the junction to

$$m\dot{a}(\tau_0^+) - m\dot{a}(\tau_0^-) = P_0 q^\sharp.$$

If P_R jumps, neither one-sided value defines P_0 without such an event assignment. Carrier-changing events likewise require a supplied map between the adjoining configuration spaces.

H Exact finite identities and numerical agreement

Finite sources permit exact evaluation of weighted-geometry and source-update identities and independent numerical realization of the same dynamics. Exact arithmetic verifies those algebraic laws; floating-point comparisons quantify agreement among relation-coordinate, spectral, first-order, and electromagnetic realizations.

H.1 Exact algebraic identities

For rational positive carrier metrics M_H and W_G , rational incidence matrices satisfy

$$D^* = W_G^{-1} D^\top M_H, \quad P_R^2 = P_R = P_R^*, \quad P_R D = D,$$

and

$$L_R = c D D^*, \quad P_R L_R = L_R, \quad \mathbb{A}_D^2 = -\text{diag}(D D^*, D^* D).$$

Rational source additions and removals also satisfy the residual and lost-subspace projector identities from [appendix B](#) exactly. Separate pairs with equal P_R and unequal L_R realize the distinction between active support and dynamical stiffness without numerical approximation.

H.2 Independent dynamical constructions

Compatible initial data evolved by the spectral second-order propagator and by the skew-adjoint first-order propagator give the same quantity trajectory. Source-free trajectories conserve positive energy and transform covariantly under clock reversal. Initial data in $\ker D^*$ evolve affinely, response jets recover P_R before L_R , and impulses applied after an observation time leave its response unchanged. Direct differentiation for a varying $D(\tau)$ reproduces the geometric-work term in [proposition 13.1](#).

Sign choice in the temporal action is also spectrally visible. Replacing the restoring equation $m\ddot{a} + L_R a = 0$ by $m\ddot{a} - L_R a = 0$ changes every positive mode λ from oscillatory motion to combinations of $e^{\pm\tau\sqrt{\lambda/m}}$. Such growth cannot generate the orthogonal finite-energy evolution required in [section 9](#).

H.3 Numerical agreement

[Table 11](#) reports maximum absolute discrepancies against the independent reference named in each row. Digital measurements use the conditions of [section 17](#); electromagnetic measurements use the finite realization of [section 18](#).

Table 11: Maximum discrepancies among independent finite realizations.

Realization	Reference quantity	Maximum discrepancy
CPU FP64 relation response	full-precision response	2.22×10^{-16}
Metal/CUDA FP32 response	CPU FP64 response	1.51×10^{-7}
CUDA FP16 relation response	CPU full-precision response	4.28×10^{-4}
Electromagnetic trajectory	decoded field trajectory	1.20×10^{-15}
Electromagnetic operator identities	pullback and intertwining	4.20×10^{-15}

Frequency, encoding, source, and readout perturbations in the electromagnetic realization produce discrepancies above 10^{-4} , separating numerical roundoff from violations of the required correspondence. Appendices [J](#) and [K](#) give the corresponding experimental constructions.

I Corpus operators and query covectors

Corpus observations give a concrete finite source in which positive empirical measure fixes the relation operator and visible local context fixes an incident covector. One covector is determined precisely when all readings retained by the observations agree after application of D . Operator construction, source covariance, and quotient by $\ker D$ establish this uniqueness criterion.

I.1 Positive empirical-measure operator

For a finite vocabulary $\mathcal{V}$, directed feature set $\mathcal{F}$, and nonnegative counts $n(w, \varphi)$, set

$$H = \ell^2(\mathcal{V}), \quad G = \ell^2(\mathcal{F}),$$

and define

$$D_{w\varphi} = \sqrt{n(w, \varphi)/N_{\text{rel}}}, \quad N_{\text{rel}} = \sum_{w, \varphi} n(w, \varphi) > 0.$$

Standard orthonormal coordinate bases give

$$\|D\|_{\text{HS}}^2 = \sum_{w,\varphi} \frac{n(w,\varphi)}{N_{\text{rel}}} = 1, \quad \|D\| \leq 1.$$

Every unit relation incident q_C therefore obeys $\|Dq_C\|_H \leq 1$. Boundedness requires neither an output normalization nor a fitted scale.

Independent row normalization instead defines

$$D_{w\varphi}^{\text{cond}} = \sqrt{\frac{n(w,\varphi)}{\sum_{\varphi'} n(w,\varphi')}}.$$

for every nonempty row. Each such row has squared norm one, so

$$\|D^{\text{cond}}\|_{\text{HS}}^2 = \# \left\{ w : \sum_{\varphi} n(w,\varphi) > 0 \right\}.$$

Conditional normalization thus removes observed term mass and replaces the joint empirical measure by one unit of mass per admitted term. This is the structural distinction evaluated in [section 16](#).

I.2 Context-induced source covectors

For nonempty directed context $C \subseteq \mathcal{F}$, set

$$q_C = |C|^{-1/2} \sum_{\varphi \in C} e_{\varphi}^G.$$

Then $\|q_C\|_G = 1$. Define

$$j_C(a) = \langle q_C, D^*a \rangle_G.$$

Cauchy–Schwarz and $\|D\| \leq 1$ give continuity. Adjointness gives

$$j_C(a) = \langle Dq_C, a \rangle_H,$$

so

$$j_C^{\sharp} = Dq_C.$$

Since $Dq_C \in \text{ran } D \subseteq B_R$,

$$P_R j_C^{\sharp} = j_C^{\sharp}.$$

Neither the concealed term nor any quantity derived from it for scoring enters q_C , j_C , or this proof, establishing [proposition 16.1](#).

I.3 Uniqueness modulo the relation null space

Let $\mathcal{I}_R(x)$ be nonempty. Suppose first that

$$D\eta_1 = D\eta_2 \quad \text{for all } \eta_1, \eta_2 \in \mathcal{I}_R(x).$$

Choosing any $\eta_0 \in \mathcal{I}_R(x)$ defines $j_x^\# = D\eta_0$, independently of the representative, and the resulting covector is faithful to every admitted reading.

Conversely, faithfulness of one covector requires $j_x^\# = D\eta$ for every $\eta \in \mathcal{I}_R(x)$. Pairwise images must therefore agree. Equivalently,

$$\mathcal{I}_R(x) \subseteq \eta_0 + \ker D.$$

This proves [theorem 7.1](#). An empty incident family records no admitted reading. A nonempty family contained in $\ker D$ instead determines the valid silent covector $j_x = 0$.

I.4 Naturality under source isomorphism

Let a source isomorphism satisfy $U_H D = D' U_G$, and suppose compatible readings transport as

$$\mathcal{I}_{R'}(x') = U_G \mathcal{I}_R(x).$$

Whenever the affine-null criterion holds,

$$j_{x'}^\# = D' U_G \eta = U_H D \eta = U_H j_x^\#.$$

Canonical covector determination is therefore independent of coordinate order and relabeling, as asserted in [proposition 7.2](#).

I.5 Distinct admitted readings

Suppose observations retain $\eta_1, \eta_2 \in \mathcal{I}_R(x)$ with

$$D\eta_1 \neq D\eta_2.$$

Any single covector would have to equal both induced covectors to remain faithful, which is impossible. Observations then determine the covector family $\mathcal{J}_R(x)$, rather than one canonical member. Additional observations may remove one reading; averaging or selecting between their distinct images does not follow from the existing source data.

I.6 Data separation in corpus evaluation

Training observations alone determine D . Validation selects between the joint-measure and conditional source constructions. Evaluation then uses 2,000 held-out occurrences that influence neither source determination nor construction selection. Visible context determines q_C , while the concealed term is used solely for scoring and never enters source or incident determination. [Appendix J](#) gives the digital response measurements.

J Computational experiments

Experiments in [section 17](#) test distinct exact reductions of the field response: relation-coordinate evaluation, reachable and observable restriction, fixed-time Green maps, matrix-free construction, source updates, component and symmetry decomposition, native processor kernels, corpus response, and event-composed propagation. Each reduced realization is compared with the unreduced response on identical admitted incident and readout coordinates.

J.1 Retrieval and FlashAttention comparison

Each retrieval record contains observed key–payload associations, a query constructed independently of its target payload, an admissibility mask, and a target used only for evaluation. Sequence length and association count were

$$(64, 4), (128, 8), (256, 16), (512, 64).$$

Prefix observations determine one relation column per admitted binding,

$$De_i^G = k_i + v_{\pi(i)}.$$

Orthogonal finite Fourier labels distinguish relation coordinates without using targets. Positive localization weights enter the finite adjoint and source construction. A query supplies $[e_i^G] \in G / \ker D$; payload reassignment changes the corresponding column of D without changing that query coordinate. Full-precision CPU evaluation of ODe_i^G provides the independent numerical reference.

For a fixed source, admitted columns of

$$\mathcal{M}_R = OD$$

determine relation-coordinate response. Incident support k and output width z give kz scalar multiply–adds by column application, while a rank- r factorization costs $r(k+z)$. Column application was selected when $kz \leq r(k+z)$. All retrieval incidents have support one, and source ranks 4, 8, 16, and 64 equal their relation dimensions. Invalid incidents return exact zero. Evaluation therefore forms no ambient query vector, explicit projector, pairwise score array, softmax, normalization pass, or weighted-value pass.

FlashAttention used one nonrecurrent FP16 scaled-dot-product-attention call over the complete prefix and query batch. Ambient query vectors encoded the same key–payload instances supplied to field response in relation coordinates. Native FlashAttention kernels excluded per-query host iteration. Relation coordinates and query, key, and value arrays are different representations, so literal input-array equality is inapplicable. Comparison uses retrieval accuracy and response time after both representations have been formed. Both realizations attained 100 percent retrieval accuracy in every condition.

Interventions permuted queries, reassigned payloads, masked invalid incidents, and withheld all targets until scoring. Every response changed or remained silent as predicted. Maximum FP16 field error relative to the full-precision relation response was 4.28×10^{-4} . Profiler traces identified both executed kernels.

Latency samples invoked device kernels after source and input-coordinate construction and host-to-device transfer. Batch and incremental results are medians of 200 synchronized evaluations after warmup. Source construction, transfer, first execution, source replacement, retained state, and peak device allocation were measured separately.

J.2 Reachable and observable reduction

For finite nonnegative self-adjoint L_R , incident map $B_{\text{in}} : U \rightarrow H$, and readout $O : H \rightarrow Z$, block Arnoldi iteration constructs the reachable carrier from the incident and declared initial-data maps. Dual closure retains its readout-visible directions through the Cayley–Hamilton degree bound. Numerical comparisons include symmetry, positivity, invariance, normalized Markov parameters, and complete response.

Examples contained eight jointly reachable and observable coordinates, five reachable but unobservable coordinates, six observable but unreachable coordinates, and a residual sector of neither kind. Enlarging incident or readout maps restored the corresponding excluded coordinates, confirming that reduction follows reachability and observability rather than accidental zero values.

Full and reduced realizations used identical incident coordinates, readout, mass, observation time, precision, and processor. Recorded quantities were ambient and retained dimensions, construction time, storage, scalar-multiply count, normalized Markov residual, response error, and median latency. Nonzero initial data enter by adjoining their position and velocity maps to the reachable construction.

J.3 Fixed-time Green response

Fixed-time examples use four-dimensional incident and readout spaces. Frequency counts 4, 16, and 64, each with residue rank four, give retained second-order dimensions 16, 64, and 256. At observation time 0.73 and mass 1.4, the visible spectral residues determine one 4×4 Green map from constant incident coordinates to observed response.

Direct modal evaluation sums every spectral contribution. Fixed-time evaluation applies the Green map once, a basis incident selects one column, and batch evaluation uses 256 device-resident incidents. Latencies are medians of seven groups of 200 repetitions after ten warmup calls. CPU uses FP64 and MPS uses FP32. Construction time, continuous-time degree, normalized moment residual, response error, operation count, storage, and scalar, basis, and batch latencies were recorded. A different observation time or forcing history determines a different response map.

J.4 Matrix-free construction and source variation

Matrix-free examples use diagonal square relation operators of ambient dimension 128, 512, and 2,048 with one nonzero entry per coordinate. Incident maps select the first eight relation coordinates, readouts observe the first four quantity coordinates, and carrier metrics are positive diagonal operators. Sparse applications of D and its metric adjoint D^* construct the minimal carrier without forming dense D or L_R . Independent dense evaluation supplies the response reference. Storage counts include sparse indices and values, carrier metrics, and active Krylov workspace; the dense comparison stores both a square whitened relation matrix and a square intrinsic operator.

Three variations test whether an existing eight-dimensional reachable carrier remains valid. One changes its first eight diagonal relation coefficients, one changes only a disconnected coefficient, and one couples the original carrier to a ninth quantity direction. Residuals of the changed incident map and intrinsic operator outside the retained carrier decide whether reconstruction may remain inside it. Independent matrix-free construction from the changed source verifies each decision and response.

Carrier-enlargement examples add one off-diagonal relation entry coupling an admitted relation coordinate to a new quantity direction that also enters the readout. Ambient dimensions are 128, 512, 2,048, and 8,192. Closure of the old basis together with the new direction under the changed factorized operator gives an augmented invariant carrier; reachability and observability are then recomputed inside it. Measurements compare added and final dimensions, factor applications, update and complete-construction times, normalized Markov residuals, and response errors.

J.5 Component and symmetry decomposition

Disconnected examples consist of 16-node incidence chains. Component counts 8, 32, 128, and 512 give ambient quantity dimensions 128, 512, 2,048, and 8,192. Incident and readout maps act only on the first component. Restriction retains every relation incident to that component and preserves the complete observed response. Independent global matrix-free construction verifies the restriction. A connected-chain control has nonzero positive-time response beyond every prescribed finite hop radius, so component removal does not imply a finite-hop cone inside a connected source.

Symmetry examples use equal diagonal relation coefficients on coordinate pairs. A signed permutation exchanges quantity and relation coordinates in each pair and defines an involutive source automorphism. Incident and readout maps contain one symmetric and one antisymmetric vector supported on every pair. Pair counts 8, 16, 32, 64, 128, and 256 give ambient dimensions 16–512. Separate parity closures reproduce the undecomposed response. Source-intertwining and port-invariance perturbations destroy the corresponding exact reduction. Construction time, factor applications, maximum block width, sector dimensions, parity residuals, and response error were measured, including sizes at which decomposition overhead exceeds its saving.

J.6 Processor-independent response maps and numerical accuracy

Digital response is represented in four forms: a static relation-coordinate map, a fixed-time Green map, a finite family of sampled-time Green maps, and an exact regular-grid recurrence. Source, incident and readout maps, admitted initial data, mass, forcing convention, and requested times or grid spacing determine each representation independently of its processor realization.

Static equality means equality with relation-coordinate response. Fixed- and sampled-time equality includes every admitted initial-position and initial-velocity coordinate. Recurrent equality includes complete observed history and final position and velocity for zero and nonzero initial states. Changing any determining datum changes the response map. Irregular forcing histories and time-dependent sources remain governed by the general dynamical law rather than by one fixed map.

CPU FP64 and MPS FP32 evaluate all four forms locally. CUDA results come only from execution on CUDA hardware. Accuracy comparisons use identical incident coordinates, initial data, and prior state and include every returned state and observed coordinate. Maximum absolute and pointwise relative errors are reported. Arithmetic-operation estimates remain separate from executed kernel counts, and a fusion claim requires a trace identifying the dispatch boundary.

J.7 Processor kernels and measured execution

Static indices, fixed- and sampled-time coordinates, and recurrent histories with prior states are transferred before response timing; their transformations are measured separately. CPU traces resolve wrapper calls into compute primitives. A fixed- or sampled-time response counts as one fused CPU operation only when its trace contains one `addmv_` and no additional response computation. Device traces synchronize before and after capture; host submission alone does not establish Metal or CUDA fusion.

Processor-specific CPU, Metal, and CUDA kernels evaluate the same response maps without redetermining the carrier or field law. Comparisons include zero-support incidents, every admitted incident and initial-data coordinate, complete recurrent histories, final position and velocity, nonzero prior states, and empty histories. Kernel source specifies the executed operations, and device traces identify the dispatched kernels. Latency includes output allocation, submission, execution, and synchronization on devices and output allocation plus execution on CPU. Tensor operations and processor-specific kernels are both retained, including workloads for which tensor execution is faster.

Modal or observation widths above one device workgroup use one parallel modal update followed by one observation kernel per step. Device-stream ordering enforces their dependency, producing twice as many launches as temporal steps; such execution is not called fused. Narrower responses use one workgroup. Tested widths begin immediately above the workgroup boundary and include two larger values.

J.8 WikiText source construction and response

Training text alone determines the sparse local-occurrence source before any held-out incident or target is selected. Sparse construction preallocates the exact number of nonzero entries and is compared entrywise with the exact count-defined sparse operator. Identity readout applies sparse source columns directly without forming a dense identity, dense relation operator, or ambient projector.

Deterministically selected held-out occurrences supply target-independent local contexts. Features absent from the training relation carrier restrict to zero without renormalizing the remaining amplitudes. Direct relation-coordinate response provides the numerical reference. Measurements separate data loading, source determination, sparse construction, held-out selection, context encoding, relation-coordinate construction, response-map construction, processor realization, execution, output transfer, and readout. Memory and operation counts include source, response, input, and output storage, retained-column and index reads, output writes, multiply-adds, device launches, synchronization, precision, and amortization over 1, 10, 100, and 1,000 batches.

One training record is then changed. Source redetermination and response-map reconstruction are compared with direct evaluation; an incremental update is used only when the residual conditions of [section 17](#) preserve the retained carrier.

J.9 Temporal propagation across constant-source intervals

Temporal conditions use 256, 1,024, and 4,096 regular steps divided into four equal constant-source intervals. Every condition has batch size 64, 64 modal coordinates, homogeneous three-coordinate modal states, and a fixed 16-coordinate final-state observation.

Regular-grid execution applies four affine modal transitions for their declared numbers of steps. Event composition applies one exact matrix power per interval. Both kernels return the complete final modal state before the common observation map, which is excluded from both timed paths. Every returned state coordinate and the declared observation are compared under maximum absolute tolerance 2×10^{-3} , and device traces identify both kernels.

Latency uses 200 synchronized samples after five warmup calls. Construction of the event powers uses 20 host samples after three warmup calls. Break-even count is the ceiling of construction time divided by per-response latency difference. Any change to an interval transition or duration determines new matrix powers. [Appendix K](#) proves the finite electromagnetic realization.

K Electromagnetic construction and calibration

Every finite positive active information-field law is a finite collection of harmonic modes. Scaled periodic cavities realize their frequencies, and a modal isometry pulls Maxwell action, energy, forcing, attention, and retarded response back to the information-field law. Modal construction and measurement calibration preserve these quantities while separating observation error from dynamical mismatch.

K.1 Transverse carrier and constraints

Let Ω be a finite disjoint union of flat periodic cavities with positive homogeneous ϵ and μ on each component. Equip vector potentials with the real L^2 pairing, impose temporal gauge, and take the mean-zero transverse carrier

$$\mathcal{T}_{\text{EM}} = \left\{ \mathbf{A} \mid \operatorname{div}(\epsilon \mathbf{A}) = 0, \int_{\Omega} \mathbf{A} = 0 \right\}.$$

On their natural transverse domains, operators

$$M_{\text{EM}} \mathbf{A} = \epsilon \mathbf{A}, \quad K_{\text{EM}} \mathbf{A} = \operatorname{curl}(\mu^{-1} \operatorname{curl} \mathbf{A})$$

are positive self-adjoint and nonnegative self-adjoint, respectively. Periodic integration by parts has no boundary remainder. With $\mathbf{E} = -\dot{\mathbf{A}}$, $\mathbf{B} = \operatorname{curl} \mathbf{A}$, transverse impressed current, and $\rho_e = 0$, Gauss and continuity constraints are

$$\operatorname{div}(\epsilon \mathbf{E}) = 0, \quad \operatorname{div} \mathbf{B} = 0, \quad \partial_{\tau} \rho_e + \operatorname{div} \mathbf{J} = 0.$$

Transversality preserves the first constraint, the identity $\operatorname{div} \operatorname{curl} = 0$ gives the second, and transverse current gives the third. Source sign is the convention fixed by the action in [section 18](#), whose Euler–Lagrange equation is $M_{\text{EM}} \ddot{\mathbf{A}} + K_{\text{EM}} \mathbf{A} = \mathbf{J}$.

K.2 Frequency scaling

Let one reference transverse eigenmode satisfy

$$K_{\text{EM}} \phi_0 = \omega_0^2 M_{\text{EM}} \phi_0.$$

Scaling every spatial length by $\ell > 0$ multiplies one derivative by ℓ^{-1} and curl–curl by ℓ^{-2} . Resulting modal frequency is

$$\omega(\ell) = \omega_0 / \ell.$$

Every desired positive frequency can therefore be realized by one scaled copy of the reference cavity.

K.3 Modal construction

Choose an H -orthonormal eigenbasis of the finite active operator,

$$L_{R,B} e_k = \lambda_k e_k, \quad \lambda_k > 0.$$

Choose the cavity scale

$$\ell_k = \omega_0 \sqrt{m / \lambda_k}$$

and normalize its selected mode by

$$\langle \phi_k, M_{\text{EM}} \phi_{\ell} \rangle = m \delta_{k\ell}.$$

Disjoint cavity supports give modal orthogonality. For $a \in B_R$, define

$$\mathcal{U}_{\text{EM}} a = \sum_k \langle e_k, a \rangle_H \phi_k$$

and set $V_{\text{EM}} = \text{ran } \mathcal{U}_{\text{EM}}$. This encoding satisfies

$$\mathcal{U}_{\text{EM}}^* M_{\text{EM}} \mathcal{U}_{\text{EM}} = m I_{B_R}$$

and

$$\mathcal{U}_{\text{EM}}^* K_{\text{EM}} \mathcal{U}_{\text{EM}} = L_{R,B}.$$

Modewise eigenvalue equations give the stronger identity

$$K_{\text{EM}} \mathcal{U}_{\text{EM}} = M_{\text{EM}} \mathcal{U}_{\text{EM}} L_{R,B} / m.$$

Applying M_{EM}^{-1} proves invariance of the selected modal range under full ideal Maxwell development, not merely under a projected equation.

K.4 Action and energy pullback

For $f \in B_R$, define the transverse current

$$J_f = M_{\text{EM}} \mathcal{U}_{\text{EM}} f / m$$

which obeys $\mathcal{U}_{\text{EM}}^* J_f = f$. Substitution into the Maxwell action gives

$$\mathcal{S}_{\text{EM}}[\mathcal{U}_{\text{EM}} a; J_f] = \int \left[\frac{m}{2} \|\dot{a}\|_H^2 - \frac{1}{2} \langle a, L_{R,B} a \rangle_H + \langle a, f \rangle_H \right] d\tau.$$

Energy pullback is

$$\frac{1}{2} \langle \mathcal{U}_{\text{EM}} \dot{a}, M_{\text{EM}} \mathcal{U}_{\text{EM}} \dot{a} \rangle + \frac{1}{2} \langle \mathcal{U}_{\text{EM}} a, K_{\text{EM}} \mathcal{U}_{\text{EM}} a \rangle = \frac{m}{2} \|\dot{a}\|_H^2 + \frac{1}{2} \langle a, L_{R,B} a \rangle_H.$$

K.5 Incident sources and retarded response

A query covector supplies

$$f_q = P_R q^\sharp, \quad J_q = J_{f_q}.$$

Pullback of static variational response changes by $-f_q$, reproducing attention. Define the modal decoder

$$C_{\text{EM}} = m^{-1} \mathcal{U}_{\text{EM}}^* M_{\text{EM}}$$

satisfies $C_{\text{EM}} \mathcal{U}_{\text{EM}} = I_{B_R}$. Let $G_{\text{EM}}^+(\tau)$ map an impulsive transverse current at zero to its zero-past vector-potential response. Strong intertwining and Cauchy uniqueness then imply, for every $f \in B_R$,

$$C_{\text{EM}} G_{\text{EM}}^+(\tau) J_f = K_R(\tau) f.$$

Instantaneous variational covector and retarded displacement remain distinct observables.

K.6 Numerical agreement and sensitivity

A finite example used $\dim H = 6$, $\dim G = 5$, active rank five, one silent quantity direction, and three unselected Maxwell modes. Across 2,401 clock samples, maximum decoded position and velocity errors remained below 1.2×10^{-15} . Pullback and intertwining residuals remained below 4.2×10^{-15} . Unselected-mode amplitude was exactly zero in ideal arithmetic.

Four perturbations alter one correspondence at a time. Stiffness perturbation produced trajectory error 1.53×10^{-3} . Mode exchange produced stiffness-pullback error 2.08×10^{-1} . Source-amplitude perturbation produced attention error 1.00×10^{-1} . Readout exchange produced output error 5.10×10^{-1} . Every error exceeded 10^{-4} .

K.7 Measurement calibration and dynamical mismatch

Let the finite selected-plus-noise modal carrier have dimension n , and let $X \in \mathbb{R}^{n \times N}$ collect calibration probes with $\text{rank } X = n$. For an invertible sensor map S_{EM} , noiseless measured probes are

$$Y = S_{\text{EM}}X.$$

Desired decoded probes are $C_{\text{EM}}X$. A decoder satisfying every probe equation obeys

$$C_{\text{EM,cal}}Y = C_{\text{EM}}X.$$

Full row rank of X and invertibility of S_{EM} give

$$C_{\text{EM,cal}} = C_{\text{EM}}S_{\text{EM}}^{-1}. \quad (\text{K.1})$$

Equation (K.1) corrects measurement coordinates only. Suppose instead that modal dynamics use $K_{\text{EM},\varepsilon}$. Exact information-field development would still require

$$K_{\text{EM},\varepsilon}\mathcal{U}_{\text{EM}} = M_{\text{EM}}\mathcal{U}_{\text{EM}}L_{R,B}/m. \quad (\text{K.2})$$

Failure of (K.2) changes the encoded trajectory before measurement. Likewise, a perturbed current must satisfy

$$\mathcal{U}_{\text{EM}}^* J_{f,\varepsilon} = f \quad (\text{K.3})$$

to retain the prescribed incident force. No observation decoder can make a trajectory violating (K.2), or a source violating (K.3), satisfy the original field equation.

A finite calculation used 384 full-rank probes, five selected modes, and three nuisance modes. Noiseless calibrated carrier error was 4.01×10^{-16} . Runtime and calibration-noise errors increased over relative levels 10^{-4} , 10^{-3} , and 10^{-2} . Stiffness and source-gain perturbations increased post-calibration trajectory errors over the same levels. Normalized modal energy and integrated source work differed by 2.45×10^{-9} . Results separate observation calibration from dynamical and source mismatch.